Set Theory Lecture Notes
MATD94/D95

1 ZFC Axioms

Definition 1A. A **set** is a collection of mathematically defined objects.

Notice this definition implies the empty collection is a set.

- An element of a set is also a set, and elements of elements are sets, too. In other words:

$$(x \text{ is a set}) \wedge (y \in x) \implies y \text{ is a set}$$
$$(x \text{ is a set}) \wedge (y \in x) \wedge (z \in y) \implies z \text{ is a set}$$
$$\vdots$$

1.1 First Order Languages

Definition 11A. **LOST** is a first order language with a binary predicate $\in$. In particular, $x \in y$ means "x is an element of y".

Definition 11B. In a **first order language**, the only symbols are:

- parentheses: (and)
- variables: v_i for any natural number i
- existence quantifier: $\exists$
  - $\forall$ is a shorthand for $\neg\exists$
- connectives: $\neg$ and $\wedge$

Definition 11C. An **expression** of a first order language is a finite sequence of symbols.

- e.g. $\exists v_0(\exists v_1(\neg$ is an expression.

Definition 11D. A **formula** is an expression constructed by the following rules:

- $v_i = v_j$ and $v_i \in v_j$ are (atomic) formulas for any natural i.
- If φ and ψ are formulas, then so are $(\varphi), (\psi), \neg(\varphi), (\varphi) \wedge (\psi), \exists v_i(\varphi)$.

Definition 11E. An occurrence of a variable x is **bound** in a formula iff x lies in the scope of quantifier $\exists$; otherwise x is **free**.

Definition 11F. A **sentence** is a formula with no free variables.

Definition 11G. An **interpretation** of LOST is defined by a nonempty domain of discourse (in which the variables are intended to vary) and a binary relation (on the said domain) as the interpretation of $\in$.

Example 11A. Consider the sentence φ defined as $\forall x \exists y (y \in x)$, which is either true or false under a given interpretation. For instance, φ is true under domain $\mathbb{Z}$ and $\in$ interpreted as $x \in y$ iff $x < y$, but φ is refutable in ZFC (namely, by the empty set).

1.2 Axioms

Axiom of Existence $\exists x (x = x)$

- A set exists.

Axiom of Extension $\forall X \forall Y (\forall z (z \in X \leftrightarrow z \in Y) \to X = Y)$

- Two sets are equal if they are comprised of the same elements.

Axiom of Pairing $\forall x \forall y \exists z (x \in z \wedge y \in z)$

- Notice such z is not necessarily $\{x, y\}$, it only contains x and y.

Axiom of Foundation $\forall x (\exists y (y \in x) \to \exists y (y \in x \wedge \nexists z (z \in x \wedge z \in y)))$

- Every nonempty set has an $\in$-minimal element.
- The axiom of foundation is also called the axiom of regularity.

Example 12A. These axioms so far imply that no set contains itself. Indeed, BWOC assume $x \in x$, we can construct the singleton $\{x\}$ by Pairing. Then, $\{x\}$ does not have a $\in$-minimal element, which contradicts Foundation.

Axiom of Union $\forall \mathcal{S}\ \exists U \forall X \forall x (x \in X \wedge X \in \mathcal{S} \to x \in U)$

- Again, notice U only contains the union of $\mathcal{S}$, not necessarily the converse.

Comprehension Schema

For any formula φ with free variables among $X, w_1, \cdots, w_n, x$

$$\forall X \forall w_1 \cdots \forall w_n \exists Y \forall x (x \in Y \leftrightarrow x \in X \wedge \varphi)$$

- We denote such Y as $\{x \in X : \varphi\}$. This axiom allows us to extract subsets of X.

Example 12B. With this axiom, we can conclude that $\mathbb{V} \stackrel{\text{def}}{=} \{x : x = x\}$ (i.e. the collection of all sets) is not a set. Indeed, if $\mathbb{V}$ were a set, we can define $D = \{x \in \mathbb{V} : x \notin x\}$ by Comprehension. However, this implies $D \in D \iff D \notin D$, a contradiction.

Replacement Schema

For any formula φ with free variables among $X, w_1, \cdots, w_n, x, y$

$$\forall X \forall w_1 \cdots \forall w_n (\forall x (x \in X \to \exists! y \varphi) \to \exists Y \forall x (x \in X \to \exists y (y \in Y \wedge \varphi)))$$

- In other words, if each $x \in X$ induces an unique y that satisfies φ, then there exists a set Y that contains all these y's. Such Y can also be viewed as a set containing the image of the mapping $x \mapsto y$ (determined by φ).

Axiom of Infinity $\exists I (0 \in I \wedge \forall x \in I \quad S(x) \in I)$

- $S(x) \stackrel{\text{def}}{=} x \cup \{x\}$ is the **successor** of x .
- This axiom states that an *inductive* set exists. With an inductive set, we can define the set of *naturals* as the smallest inductive set.

Axiom of Power Set $\forall X \exists \mathcal{S}\ \forall A (A \subseteq X \to A \in \mathcal{S})$

- $A \subseteq X$ is abbreviation for $\forall a (a \in A \to a \in X)$

Axiom of Choice $\forall X \exists R (R \text{ well orders } X)$

- Notice this is strictly stronger than "every nonempty family has a choice function." A *choice function* of some nonempty family $\mathcal{S}$ is $g : \mathcal{S} \to \bigcup \mathcal{S}$ such that
$$\forall X \in \mathcal{S} \quad g(X) \in X.$$

With all the axioms, we can define the following sets.

Definition 12A. For any set $\mathcal{S}$, let U be a set as per Axiom of Union. The **union** of $\mathcal{S}$ is the set

$$\bigcup \mathcal{S} \stackrel{\text{def}}{=} \{x \in U : \exists S \in \mathcal{S} \quad x \in S\}.$$

Definition 12B. For any set $\mathcal{S}$ with some $S_0 \in \mathcal{S}$ (i.e. $\mathcal{S}$ is nonempty), the **intersection** of $\mathcal{S}$ is the set

$$\bigcap \mathcal{S} \stackrel{\text{def}}{=} \{x \in S_0 : \forall S \in \mathcal{S} \quad x \in S\}.$$

- Note that $\mathcal{S}$ has to be nonempty, otherwise we have that $\bigcap \varnothing = \mathbb{V}$.

Definition 12C. For sets X and Y, the **cartesian product** of X and Y is the set:

$$X \times Y = \{(x, y) : x \in X \wedge y \in Y\}$$

- where $(x, y) \stackrel{\text{def}}{=} \{x, \{x, y\}\}$

Proposition 12A. $X \times Y$ is a set.

Proof.

Fix $x \in X$, then for any $y \in Y$, the set $(x, y) \stackrel{\text{def}}{=} \{x, \{x, y\}\}$ is unique. Therfore, by Replacement and Comprehension, we have that

$$\{(x,y)\}_{y\in Y} = \{x\} \times Y$$

is a set.

Notice that for any $x \in X$, the set $\{x\} \times Y$ is also unique. So, by Replacement and Comprehension again, we get the set $\{\{x\} \times Y\}_{x\in X}$ whose union is exactly $X \times Y$. $\square$

Exercises

Exercise 1.1. (Group X) Determine if the following statement is stronger or equivalent to Axiom of Choice:

- For any relation R on a nonempty X such that $\forall a \in X\ \exists b \in X\ aRb$, then
$$\exists f : \omega \to X \text{ such that } \forall i \in \omega \quad f(i)Rf(i+1)$$

Exercise 1.2. (Group Y) Determine if the following statement is stronger or equivalent to Axiom of Choice:

- For any set X, there exists a linear order R on X such that
$$\forall A \subseteq X \quad \exists z \in A \quad \forall y \neq z \in A \quad y \not R z$$

Exercise 1.3. (Group Y) Given sets x and y, clearly use ZFC axioms to formulate the intersection of x and y.

2 Ordinal Numbers

2.1 Relations, Functions, Orders

Definition 21A. A **relation** between set X and set Y is a subset $R \subseteq X \times Y$.

- xRy denotes $(x,y) \in R$, and $x \not R y$ denotes $(x,y) \notin R$.

Definition 21B. The **domain** of a relation R is

$$\operatorname{dom} R \stackrel{\text{def}}{=} \{x \in X : \exists y(xRy)\}.$$

Definition 21C. The **range** of a relation R is

$$\operatorname{ran} R \stackrel{\text{def}}{=} \{y \in Y : \exists x(xRy)\}.$$

Definition 21D. A **function** on X into Y is a relation $f \subseteq X \times Y$ where each $x \in \operatorname{dom} f$ has an unique $y \in Y$ such that $(x,y) \in f$.

- Since such y is uniquely determined by x, we denote it by $f(x)$.

We also use $f : X \to Y$ to mean that domain of f is X, and the range of f is a subset of Y.

Definition 21E. A (strict) **total/linear order** on set X is a relation $R \subseteq X \times X$ satisfying

- *irreflexivity*: $\forall x \in X \quad x \not R x$,
- *transitivity*: $\forall x, y, z \in X \quad xRy \wedge yRz \implies xRz$,
- *trichotomy*: $\forall x, y \in X \quad x = y \vee xRy \vee yRx$.

Definition 21F. $y \in X$ is R-**minimal** element in X iff $\nexists x(x \in X \wedge xRy)$.

Definition 21G. Two linearly order sets (losets) $(A, <_A)$ and $(B, <_B)$ are **order isomorphic**, denoted as $(A, <_A) \cong (B, <_B)$, iff there exists $f : A \xrightarrow{\sim} B$ such that for any $x, y \in A$

$$x <_A y \iff f(x) <_B f(y).$$

We will write $A \cong B$ to denote that A is order isomorphic to B when the order is clear from context.

2.2 Well-Ordered Sets

Definition 22A. A linear order R on X is a **well-order** iff any nonempty $A \subseteq X$ has a R-minimal element.

- The **initial segment** of X given by $a \in X$ is defined as the set of predecessors of a

$$\operatorname{pred}(X, a, R) \stackrel{\text{def}}{=} \{x \in X : xRa\}.$$

Lemma 22A. No well-ordered set (woset) is order isomorphic to an initial segment of itself.

Proof. BWOC assume $f : W \xrightarrow{\sim} \operatorname{pred}(W, a, <)$ is an order isomorphism for some woset $(W, <)$ and $a \in W$. Thus being an element of the initial segment, we have that $f(a) < a$. Therefore $A = \{x \in W : f(x) \neq x\}$ is a nonempty subset of W, so let t be its minimum element.

By the trichotomy of linear orders, $f(t) < t$ or $t < f(t)$. In the first case, we find that $f(f(t)) < f(t)$ as f is an order isomorphism. Therefore $f(t) \in A$ which contradicts that t is minimal.

The other case, when $t < f(t)$, is left as Exercise 2.1. □

Theorem 22B. For wosets $(W_1, <_1)$ and $(W_2, <_2)$, exactly one satisfies:

(1) $W_1 \cong W_2$,;
(2) $W_1 \cong \operatorname{pred}(W_2, a_2, <_2)$ for some $a_2 \in W_2$;
(3) $\operatorname{pred}(W_1, a_1, <_1) \cong W_2$ for some $a_1 \in W_1$.

In any case, the order isomorphism is unique.

Proof. Let relation $f \subseteq W_1 \times W_2$ be defined as

$$f = \{(a, b) : \operatorname{pred}(W_1, a, <_1) \cong \operatorname{pred}(W_2, b, <_2)\}.$$

The rest is left as Exercise 2.2. □

2.3 Ordinal Numbers

Definition 23A. A set α is an **ordinal** iff it is
- transitive: every element of α is a subset of α,
- well-ordered by $\in_\alpha = \{(a, b) \in \alpha \times \alpha : a \in b\}$.

Definition 23B. For ordinals α and β, we define the liner order $<$ where

$$\alpha < \beta \iff \alpha \in \beta$$

Similarly, $\alpha \leqslant \beta$ iff $\alpha < \beta$ or $\alpha = \beta$.

Example 23A. $\{\{\{\varnothing\}\}, \{\varnothing\}, \varnothing\}$ is not an ordinal since its not a well-ordered by $\in$ (in particular, the order is not transitive) even though every nonempty subset has a $\in$-minimal element.

Theorem 23A. Let α, β, γ be ordinals.
(I) If $x \in \alpha$, then x is an ordinal and $x = \{\beta \in \alpha : \beta < x\}$.
(II) If $\alpha \cong \beta$, then $\alpha = \beta$.
(III) Either $\alpha = \beta$ or $\alpha < \beta$ or $\alpha > \beta$.
(IV) If $\alpha \leqslant \beta$ and $\beta \leqslant \gamma$, then $\alpha \leqslant \gamma$.
(V) If C is a nonempty set of ordinals, then C has a $\in$-minimal element.

Proof. (Theorem 23A III) If $\alpha \cong \beta$, then $\alpha = \beta$ as wanted (Theorem 23A II). Otherwise one is a initial segment of another since α and β are wosets. So WLOG assume $\alpha \cong \operatorname{pred}(\beta, b, <)$ for some $b \in \beta$. Thus $\alpha = b < \beta$ as wanted □

Proof. (Theorem 23A V) We first note that a nonempty set C of ordinals has an $\in$-minimal element iff

$$\exists z \in C \quad z \cap C = \varnothing.$$

So choose an $\alpha \in C$ and WLOG assume $\alpha \cap C$ is nonempty. Thus, $\alpha \cap C$ has a minimal element in α, which is also minimal in C. □

Note that Theorem 23A V implies the following lemma:

Lemma 23B. Any transitive set of ordinals is an ordinal.

Finally, we conclude section Section 2.3 with two important theorems.

Theorem 23C. The collection of all ordinals, $\mathbb{ON}$, is not a set.

Proof. If $\mathbb{ON}$ is a set, then it is transitive since each $\alpha \in \mathbb{ON}$ consists of ordinals (Theorem 23A I), so $\alpha \subseteq \mathbb{ON}$. This establishes that $\mathbb{ON}$ itself is an ordinal, so $\mathbb{ON} \in \mathbb{ON}$ which leads to a contradiction. □

Theorem 23D. Every woset is order isomorphic to an unique ordinal.

Proof. Consider any woset (W, R), we will show W is order isomorhic to an ordinal (the uniqueness follows from Theorem 23A II). First, define

$$A = \{a \in W : \exists \alpha_a \in \mathbb{ON} \quad \operatorname{pred}(W, a, R) \cong \alpha_a\}.$$

By Theorem 23A II, each α_a is uniquely determined by $a \in W$. Thus let f be the map $a \mapsto \alpha_a$, we have that A is order isomorphic to $f[A]$. Therefore, it is sufficient to show that $A = W$ and $f[A]$ is transitive (thus $f[A]$ is an ordinal by Lemma 23B).

To show $f[A]$ is transitive, consider any $\alpha_a \in f[A]$ and let any $\beta \in \alpha$, and we will show $\beta \in f[A]$. Let $\varphi : \operatorname{pred}(W, a, R) \xrightarrow{\sim} \alpha_a$ be an order isomorphism, and let $b = \varphi^{-1}(\beta)$. Therefore, $\operatorname{pred}(W, b, R)$ is order isomorphic to β by $\varphi \restriction \operatorname{pred}(W, b, R)$. So $a \in A$ and $\beta = f(b) \in f[A]$.

Finally, BWOC assume $A \neq W$, then by Theorem 22B, $A = \operatorname{pred}(W, c, R)$ for some $c \in W$. However, this means that $\operatorname{pred}(W, c, R) \cong f[A]$ by f, so $c \in A = \operatorname{pred}(W, c, R)$, meaning $c < c$, a contradiction. □

Definition 23C. For any woset (W, R), the **order type** of W, denoted as $\operatorname{otp}(W, R)$, is the unique ordinal that is order isomorphic to (W, R).

2.4 Ordinal Addition

Remark 24A. If X is a nonempty set of ordinals, then

$$\sup X \stackrel{\text{def}}{=} \bigcup X,$$
$$\min X \stackrel{\text{def}}{=} \bigcap X$$

are ordinals as well.

Definition 24A. The **successor** of an ordinal α is the ordinal

$$S(\alpha) \stackrel{\text{def}}{=} \alpha \cup \{\alpha\}.$$

Remark 24B. $\beta < S(\alpha)$ iff $\beta \leqslant \alpha$.

Definition 24B. An ordinal α is a **successor ordinal** iff α is the successor of some ordinal. A nonzero ordinal that is not a successor ordinal is a **limit ordinal**.

Remark 24C. If α is a limit ordinal, then $\alpha = \sup_{\beta<\alpha} \beta$.

Definition 24C. If α is a limit ordinal, then $\alpha - 1 \stackrel{\text{def}}{=} \alpha$. Othewise $\alpha - 1 \stackrel{\text{def}}{=} \beta$ where $\alpha = S(\beta)$

Definition 24D. The **sum** of two ordinals α and β (denoted as $\alpha + \beta$) is $\operatorname{otp}(W, R)$ where

$$\begin{aligned} W &= \alpha \times \{0\} \cup \beta \times \{1\} \\ (\zeta, 0) R (\eta, 0) &\iff \zeta < \eta \\ (\zeta, 0) R (\eta', 1) &\iff \zeta \in \alpha \wedge \eta' \in \beta \\ (\zeta', 1) R (\eta', 1) &\iff \zeta' < \eta' \end{aligned}$$

Notice $(\alpha \times \{0\}) \times (\beta \times \{1\}) \subset R$.

The sum $\alpha + \beta$ can be visualised as stacking β on top of α, with the order being left to right, then bottom to top.

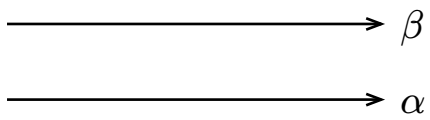


Figure 1: Stacking β on top of α

Example 24A. $\omega = 2 + \omega \neq \omega + 2$.

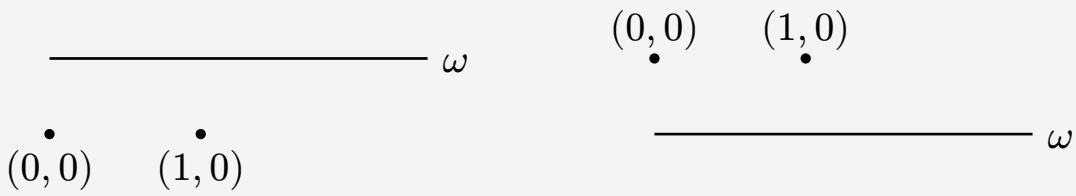


Figure 2: Visualisation of $2 + \omega$ (left) vs $\omega + 2$ (right)

Theorem 24A. For ordinals α, β, γ

$$\begin{aligned} (\alpha + \beta) + \gamma &= \alpha + (\beta + \gamma) \\ \alpha + 0 &= \alpha \\ \alpha + 1 &= S(\alpha) \\ \alpha + S(\beta) &= S(\alpha + \beta) \end{aligned}$$

If β is a limit ordinal, then

$$\alpha + \beta = \sup\{\alpha + \eta : \eta < \beta\}.$$

2.5 Ordinal Multiplication

Definition 25A. The **product** of two ordinals α and β (denoted as $\alpha \cdot \beta$) is $\operatorname{otp}(\beta \times \alpha, \leqslant_{\text{lex}})$ where $\leqslant_{\text{lex}}$ is the lexicographic order:

$$(\zeta, \eta) \leqslant_{\text{lex}} (\zeta', \eta') \iff (\zeta < \zeta') \text{ or } (\zeta = \zeta' \text{ and } \eta \leqslant \eta').$$

The product $\alpha \cdot \beta$ can be visualised as taking β copies of α, with the order being left to right, bottom to top.

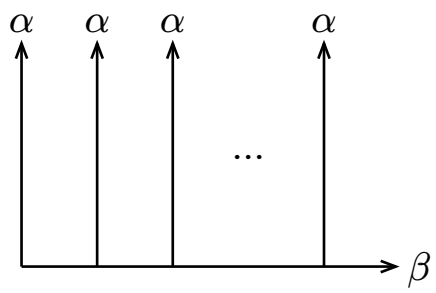


Figure 3: β copies of α

Example 25A. $2 \cdot \omega = \omega$ and $\omega \cdot 2 = \omega + \omega$.

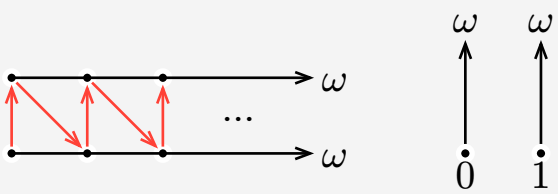


Figure 4: Visualisation of $2 \cdot \omega$ (left) vs $\omega \cdot 2$ (right)

Theorem 25A. For ordinals α, β, γ

$$(\alpha \cdot \beta) \cdot \gamma = \alpha(\beta \cdot \gamma)$$
$$\alpha \cdot 0 = 0$$
$$\alpha \cdot 1 = \alpha$$
$$\alpha \cdot (\beta + 1) = \alpha \cdot \beta + \alpha$$
$$\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$$

Example 25B. $\alpha \cdot (\beta + 1) = \alpha \cdot \beta + \alpha$.

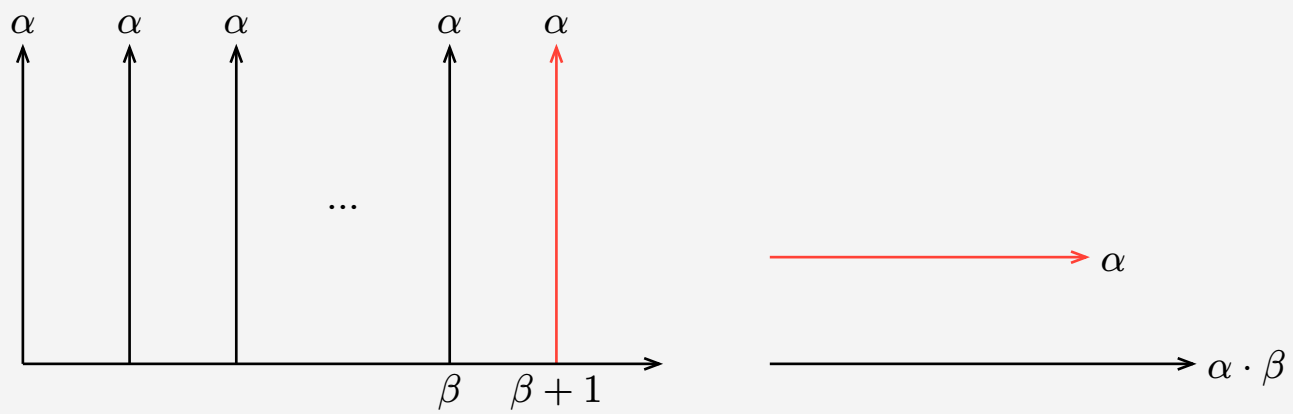


Figure 5: $\alpha \cdot (\beta + 1)$ (left) is order isomorphic to $\alpha \cdot \beta + \alpha$ (right)

Example 25C. $\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$.

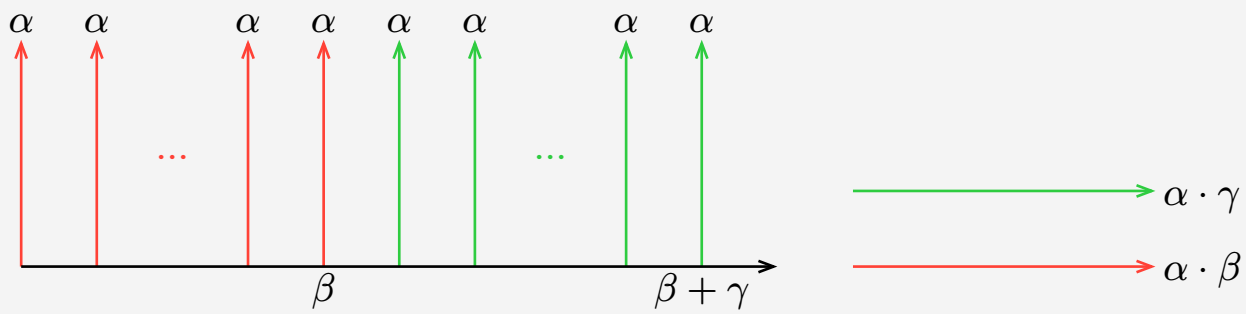


Figure 6: $\alpha \cdot (\beta + \gamma)$ (left) is order isomorphic to $\alpha \cdot \beta + \alpha \cdot \gamma$ (right)

2.6 Transfinite Induction

Theorem 26A. (Transfinite Induction on $\mathbb{ON}$) Let $C \subseteq \mathbb{ON}$ be a nonempty subclass of ordinals. Then, C has a least element.

Notice the difference between this theorem and Theorem 23A V.

A proof by (transfinite) induction on ordinal α is a proof of some statement $\varphi(\alpha)$, assuming $\varphi(\beta)$ for any $\beta < \alpha$. This technique allows us to define a function on α recursively using information available about the function below α.

Theorem 26B. (Transfinite Recursion) For any $F : \mathbb{V} \to \mathbb{V}$, there exists a unqiue $G : \mathbb{ON} \to \mathbb{V}$ such that

$$\forall \alpha \in \mathbb{ON} \quad G(\alpha) = F(G \restriction \alpha)$$

Proof.

To show existence, we define $g : \delta \to \operatorname{ran} g$ to be an δ-*approximation* of F iff

$$\forall \varepsilon < \delta \quad g(\varepsilon) = F(g \restriction \varepsilon)$$

- Exercise 2.4 (1): using induction, show that if g is a δ-approximation, and g' is a δ'-approximation, then $g \restriction (\delta \cap \delta') = g' \restriction (\delta \cap \delta')$

- Exercise 2.4 (2): using induction, show that $\forall \delta \in \mathbb{ON} \quad \exists! \delta$-approximation of F
- Exercise 2.4 (3): let $G(\alpha) = g(\alpha)$ where g is a δ-approximation of F for some $\delta > 0$, show that $G(\alpha) = F(G \restriction \alpha)$ for any $\alpha \in \mathbb{ON}$

To show uniqueness, consider any $G_1, G_2 : \mathbb{ON} \to \mathbb{V}$

- Exercise 2.4 (4): using induction on $\alpha \in \mathbb{ON}$, show that $G_1(\alpha) = G_2(\alpha)$ $\square$

Remark 26A. Informally, we can define ordinal addition using recursion as follows:

$$\begin{aligned} \alpha + 0 &\stackrel{\text{def}}{=} \alpha, \\ \alpha + S(\beta) &\stackrel{\text{def}}{=} S(\alpha + \beta), \\ \alpha + \beta &\stackrel{\text{def}}{=} \sup_{\gamma < \beta} \alpha + \gamma \quad \beta \text{ is a limit ordinal.} \end{aligned}$$

2.7 Ordinal Exponentiation

Definition 27A. For any set A and B, the set of functions from A to B is

$$^{A}B \stackrel{\text{def}}{=} \{f \subseteq A \times B | f : A \to B\}$$

The set of finite *partial* functions from ω to A is

$$A^{<\omega} \stackrel{\text{def}}{=} \bigcup_{n \in \omega} {}^{n}A$$

Definition 27B. For functions f, g with ordinal domains α, β repsectively, the **concatenation** of f and g is $f^\frown g : \alpha + \beta \to \operatorname{ran} f \cup \operatorname{ran} g$ where

$$\begin{aligned} f^\frown g \restriction \alpha &= f, \\ f^\frown g(\alpha + \gamma) &= g(\gamma) \quad \forall \gamma \in \beta. \end{aligned}$$

Definition 27C. By recursion, **ordinal exponentiation** is defined as

$$\begin{aligned} \alpha^0 &\stackrel{\text{def}}{=} 1 \\ \alpha^{S(\beta)} &\stackrel{\text{def}}{=} \alpha^\beta \cdot \alpha \\ \alpha^\beta &\stackrel{\text{def}}{=} \sup_{\gamma < \beta} \alpha^\gamma \quad \beta \text{ is a limit ordinal} \end{aligned}$$

Exercises

Exercise 2.1. (Group Y) Finish the proof of Lemma 22A.

Exercise 2.2. (Group Y) Finish the proof of Theorem 22B.

Exercise 2.3. (Group Y) Prove Theorem 23A I.

Exercise 2.4. (Group Y) Finish the proof of Theorem 26B

3 Cardinal Numbers

Definition 3A. If set A is well-orderable, then the **cardinality** of A is

$$|A| \stackrel{\text{def}}{=} \min\{\alpha \in \mathbb{ON} : \exists \varphi : \alpha \xrightarrow{\sim} A\}.$$

An ordinal α is a **cardinal** iff $|\alpha| = \alpha$.

- i.e. $\forall \beta < \alpha \quad \nexists \varphi : \alpha \hookrightarrow \beta$,
- i.e. $\forall \beta < \alpha \quad \nexists \psi : \beta \twoheadrightarrow \alpha$.

Corollary 3A.

- $|\alpha| \leqslant \alpha$ for any ordinal α.
- $n \in \omega$ is a cardinal.
- ω is a cardinal.

Theorem 3B. (Cantor-Berstein) For well-orderable sets A and B

$$\text{If } \exists \varphi : A \hookrightarrow B \text{ and } \exists \psi : B \hookrightarrow A, \text{then } |A| = |B|.$$

Lemma 3C. For ordinals α and β, if $|\alpha| \leqslant \beta \leqslant \alpha$, then $|\alpha| = |\beta|$

Proof. Let $f : \alpha \xrightarrow{\sim} |\alpha|$. Let $i_1 : |\alpha| \hookrightarrow \beta$ and $i_2 : \beta \hookrightarrow \alpha$ be inclusion maps. Thus by Cantor Berstein with $i_1 \circ f : \alpha \hookrightarrow \beta$ and $i_2 : \beta \hookrightarrow \alpha$, we have that $|\alpha| = |\beta|$ □

Corollary 3D. For ordinal α and cardinal κ, if $|\alpha| < \kappa$, then $\alpha < \kappa$.

Proof. BWOC assume $\kappa \leqslant \alpha$, then $|\alpha| = |\kappa| = \kappa$, contradicting $|\alpha| < \kappa$. □

3.1 Cardinal Arithmetics

Definition 31A. For cardinals κ and λ

$$\kappa \boxplus \lambda \stackrel{\text{def}}{=} |\kappa \times \{0\} \cup \lambda \times \{1\}|$$
$$k \otimes \lambda \stackrel{\text{def}}{=} |\kappa \times \lambda|$$

Remark 31A.

- $\boxplus$ and $\otimes$ are commutative.
- $|\kappa + \lambda| = \kappa \boxplus \lambda = |\lambda + \kappa|$.
- $|\omega + 1| = \omega$.

Lemma 31A. For any $n, m \in \omega$

$$m \boxplus n = m + n < \omega$$
$$m \otimes n = m \cdot n < \omega$$

Theorem 31B. Every infinite cardinal is a limit ordinal

Proof. BWOC assume some infinite cardinal $\kappa = \alpha + 1$ for some ordinal α. Thus $|1 + \alpha| = |\alpha|$ since α is infinite. Therefore, we have

$$\kappa = |\kappa| = |\alpha + 1| = |1 + \alpha| = |\alpha|.$$

Thus $\kappa = |\alpha| \leqslant \alpha$, contradicting that $\kappa = \alpha + 1 > \alpha$. □

Theorem 31C. For any infinite cardinal κ

$$\kappa \otimes \kappa = \kappa.$$

Proof. An injection from κ to $\kappa \times \kappa$ is straight forward, so we are left to show that $\kappa \otimes \kappa \leqslant \kappa$.

By induction, assume $\alpha \otimes \alpha = \alpha$ for any cardinal $\alpha < \kappa$. We first define a well-order $<_{\mathrm{ML}}$ on $\kappa \times \kappa$ as follows:

$$\begin{aligned}(\alpha_1, \alpha_2) <_{\mathrm{ML}} (\alpha_1', \alpha_2') \iff \max\{\alpha_1, \alpha_2\} < \max\{\alpha', \beta'\} \text{ or} \\ \max\{\alpha_1, \alpha_2\} = \max\{\alpha_1, \alpha_2\} \text{ and } (\alpha_1, \alpha_2) <_{\mathrm{lex}} (\alpha_1', \alpha_2')\end{aligned}$$

With this well-order (why not just $<_{\mathrm{lex}}$?), we will show

$$|\kappa \times \kappa| \leqslant \operatorname{otp}(\kappa \times \kappa, <_{\mathrm{ML}}) \leqslant \sup_{\delta < \kappa} \operatorname{otp}(\delta \times \delta, <_{\mathrm{ML}}) \leqslant \kappa.$$

The left-most inequality simply follows from the definition of cardinality. Next, we will show that $\operatorname{otp}(\kappa \times \kappa) \leqslant \sup_{\delta<\kappa} \operatorname{otp}(\delta \times \delta)$.

Let any $\alpha < \operatorname{otp}(\kappa \times \kappa)$, we claim that $\alpha < \operatorname{otp}(\delta \times \delta)$ for some $\delta \in \kappa$. Indeed, let S be an initial segment of $\kappa \times \kappa$ given by some (α_1, α_2) that is isomorphic to α. Define $\delta = \max\{\alpha_1, \alpha_2\} + 1 < \kappa$ (notice κ is a limit ordinal). Thus for any $(\beta_1, \beta_2) <_{\mathrm{ML}} (\alpha_1, \alpha_2)$,

$$\max\{\beta_1, \beta_2\} \leqslant \max\{\alpha_1, \alpha_2\} < \delta.$$

This implies that $(\beta_1, \beta_2) \in \delta \times \delta$, thus S is also an initial segment of $\delta \times \delta$. Therefore $\alpha < \operatorname{otp}(\delta \times \delta)$ as wanted.

To show $\sup_{\delta<\kappa} \operatorname{otp}(\delta \times \delta) \leqslant \kappa$, consider any $\delta < \kappa$ and apply the induction hypothesis:

$$|\operatorname{otp}(\delta \times \delta)| = |\delta \times \delta| = |\delta| \leqslant \delta < \kappa.$$

Therefore $\operatorname{otp}(\delta \times \delta) < \kappa$ by Corollary 3D. □

Corollary 31D. For any infinite cardinal κ and λ

(I) $\kappa \boxplus \lambda = \kappa \otimes \lambda = \max\{\kappa, \lambda\}$

(II) $|\kappa^{<\omega}| = \kappa$

Proof. (II) An injection from κ to $\kappa^{<\omega}$ is straight forward. The other direction is left as Exercise 3.2. Here is an outline:

1. For any $n \in \omega$, construct $\varphi_n : \kappa^n \hookrightarrow \kappa$ by induction.
2. Construct $\varphi : \kappa^{<\omega} \hookrightarrow \omega \times \kappa$ using φ_n.
3. Conclude $\kappa^{<\omega} \leqslant \kappa$ using Corollary 31D (I) □

Theorem 31E. (Cantor's Theorem) For any set X

$$|X| < |\mathcal{P}(X)|.$$

Proof. BWOC assume $|\mathcal{P}(X)| \leqslant |X|$, that is, $\exists \varphi : \mathcal{P}(X) \hookrightarrow X$. For any $a \in \operatorname{ran} \varphi$, let $A_a = \varphi^{-1}(a)$. Define $E = \{a \in \operatorname{ran} \varphi : a \notin A_a\}$, thus $\exists e \in X$ such that $E = A_e$.

Notice that if $e \in E$, then $e \notin A_e = E$, but if $e \notin E$, then $e \in A_e = E$, so we have a contradiction in any case. □

Corollary 31F. For every ordinal α, there exists a cardinal κ such that $\kappa > \alpha$.

Proof. Consider $W = \{R : R \text{ well-orders } \alpha \times \alpha\}$, let $S = \{\operatorname{otp}(\alpha, R) : R \in W\}$.

- Exercise 3.3: show that $\sup S$ is a cardinal larger than α. □

3.2 Cardinals

Definition 32A. The **successor** of cardinal κ is κ^+, the least cardinal greater than κ.

- A cardinal is a **successor** cardinal iff it's the successor of some cardinal, otherwise it's a **limit** cardinal

Definition 32B. By recursion, define $\aleph_\alpha = \omega_\alpha$ for any ordinal α:

$$\aleph_0 = \omega$$

$$\aleph_{S(\alpha)} = \omega_\alpha^+$$

If α is a limit ordinal,

$$\aleph_\alpha = \sup_{\beta<\alpha} \aleph_\beta.$$

Lemma 32A. For ordinals α and β,

1. $\aleph_\alpha$ is a cardinal for any ordinal α.
2. Every infinite cardinal is in the form $\aleph_\alpha$ for some ordinal α.
3. $\alpha < \beta \implies \omega_\alpha < \omega_\beta$.
4. α is a limit ordinal $\iff \aleph_\alpha$ is a limit cardinal.
   i.e. α is a successor ordinal $\iff \aleph_\alpha$ is a successor cardinal.

Lemma 32B. (AC) If κ is an infinite cardinal and $\{X_\alpha : \alpha < \kappa\}$ is a family of sets where $|X_\alpha| \leqslant \kappa$ for any α, then

$$\left|\bigcup_{\alpha<\kappa} X_\alpha\right| \leqslant \kappa.$$

Proof.

Choose $\varphi_\alpha : X_\alpha \hookrightarrow \kappa$ by AC; we will construct $\varphi : \bigcup_\alpha X_\alpha \hookrightarrow \kappa \times \kappa$ and apply Theorem 31C.

Let $X = \bigcup_\alpha X_\alpha$, and for any $x \in X$, define $\alpha_x = \min\{\alpha \in \kappa : x \in X_\alpha\}$. Finally, let $\varphi(x) = \left(\alpha_x, \varphi_{\alpha_x}(x)\right)$ which we will show is injective.

Consider any $x, x' \in X$ and assume $(\alpha, \varphi_\alpha(x)) = (\alpha, \varphi_\alpha(x'))$ for $\alpha = \alpha_x$. Then we have $\varphi_\alpha(x) = \varphi_\alpha(x')$, so $x = x'$ since φ_α is injective. □

3.3 Löwenheim–Skolem theorem

Definition 33A. For a set A

- a **0-ary** function on A is $f \in A$;
- for $n > 0$, an **n-ary** function on A is $f : A^n \to A$.

Definition 33B. For an n-ary $f : A^n \to A$, a subset $B \subseteq A$ is **closed** under f iff $f[B^n] \subseteq B$.

Definition 33C. For a set $\mathcal{L}$ of n-ary functions on A, the **closure** of $B \subseteq A$ under $\mathcal{L}$ is the minimal set $C \subseteq A$ such that

1. $B \subseteq C$, and
2. C is closed under f for any $f \in \mathcal{L}$.

More precisely, let $\mathcal{C} = \{C \subseteq A : B \subseteq C \text{ and } \forall f \in \mathcal{L} \quad f[C^n] \subseteq C\}$, then the closure of B is $\bigcap \mathcal{C}$.

Theorem 33A. (Löwenheim–Skolem theorem) Let κ be an infinite cardinal, and $\mathcal{L}$ be a family of finite-ary function of A with $|\mathcal{L}| \leqslant \kappa$, then for any $B \in [A]^{\leqslant\kappa}$, the closure of B under $\mathcal{L}$ has cardinality at most κ.

Proof.

We first define C_k recursively:

$$C_0 = B$$
$$C_{k+1} = C_k \cup \bigcup_{f \in \mathcal{L}} f[C_k^n]$$

Now, since $|C_0| \leqslant \kappa$ and $|f[C_k^n]| \leqslant \kappa$, we have that $|C_{k+1}| \leqslant \kappa$ by Lemma 32B. Finally, notice that $C = \sup_{k<\omega} C_k$ is exactly the closure of B under $\mathcal{L}$ and $|C| \leqslant \kappa$ by Lemma 32B again. □

Example 33A. (Group Theory) Let $(G, \cdot)$ be an infinite group, then G has an countable subgroup. Indeed, choose a countable subset $B \subseteq G$ and apply Theorem 33A on $\mathcal{L} = \{(a, b) \mapsto a \cdot b, a \mapsto a^{-1}\}$. Thus the closure of B under $\mathcal{L}$ is an countable subgroup of G.

3.4 Cardinal Exponentiation

Definition 34A. (AC) the **exponentiation** of cardinal κ and λ is

$$\kappa^\lambda \stackrel{\text{def}}{=} |{}^\lambda\kappa|$$

Lemma 34A. if λ is an infinite cardinal and $2 \leqslant \kappa \leqslant \lambda$, then

$$2^\lambda = \kappa^\lambda = \lambda^\lambda = |\mathcal{P}(\lambda)|.$$

Proof. First observe that

$$2^\lambda \leqslant \kappa^\lambda \leqslant \lambda^\lambda \leqslant |\mathcal{P}(\lambda \times \lambda)|.$$

Next, by Theorem 31C we have that $|\mathcal{P}(\lambda \times \lambda)| = |\mathcal{P}(\lambda)|$, and $|\mathcal{P}(\lambda)| = 2^\lambda$ is established by the function $\varphi : 2^\lambda \xrightarrow{\sim} \mathcal{P}(\lambda)$ defined as

$$\varphi(f) = \{x \in \lambda : f(x) = 1\}.$$

□

Lemma 34B. For cardinals κ, λ, μ

$$\kappa^{\lambda \boxplus \mu} = \kappa^\lambda \otimes \kappa^\mu$$
$$\left(\kappa^\lambda\right)^\mu = \kappa^{\lambda \otimes \mu}$$

Exercises

Exercise 3.1. (Group Y) Prove Corollary 31D (I).

Exercise 3.2. (Group Y) Finish the proof of Corollary 31D (II).

Exercise 3.3. (Group Y) Finish the proof of Corollary 31F.

Exercise 3.4. (Group Y) Explain why $\leqslant_{\mathrm{ML}}$ is used instead of simply $\leqslant_{\mathrm{lex}}$ in the proof of Theorem 31C.

Exercise 3.5. (Group Y) Explain where AC is used in the proof of Lemma 32B.

Exercise 3.6. (Group Y) Prove Lemma 34B.

4 Cofinality

4.1 Cofinality of Ordinals

Definition 41A. A subset A of ordinal α is a **cofinal** subset iff A is unbounded in α. In symbols:

$$\forall \beta \in \alpha \quad \exists a \in A \quad \beta \leqslant a.$$

Definition 41B. For ordinals α and β, a function $f : \alpha \to \beta$ maps α **cofinally** into β iff the range, $f[\alpha]$, is a cofinal subset of β.

Definition 41C. The **cofinality** of ordinal α (denoted $\operatorname{cof} \alpha$) is the minimum ordinal from which there exists a function that maps cofinally into α.

$$\operatorname{cof} \alpha = \min\{\beta \leqslant \alpha : \exists f : \beta \to \alpha \text{ cofinal}\}.$$

Lemma 41A. For any ordinal α

(I) $\operatorname{cof} \alpha = \min\{|A| : A \subseteq \alpha \text{ cofinal}\}$.

(II) $\operatorname{cof} \alpha \leqslant \alpha$.

(III) There exists a strictly increasing cofinal map from $\operatorname{cof} \alpha$ into α.

Proof. Lemma 41A (III)

Let $g : \operatorname{cof} \alpha \to \alpha$ be a cofinal map as per definition of $\operatorname{cof} \alpha$. We proceed to construct an strictly increasing cofinal map $f : \operatorname{cof} \alpha \to \alpha$ by recursion.

Fix $\beta \in \operatorname{cof} \alpha$ and assume $f(\gamma)$ has been defined for any $\gamma < \beta$. Define

$$f(\beta) = \max\{g(\beta), \sup_{\gamma<\beta} f(\gamma) + 1\}.$$

Taking the maximum ensures that f is increasing and cofinal. In particular, if we were to define $f'(\beta) = \sup_{\gamma<\beta} f'(\gamma) + 1$, then its range might be bounded. □

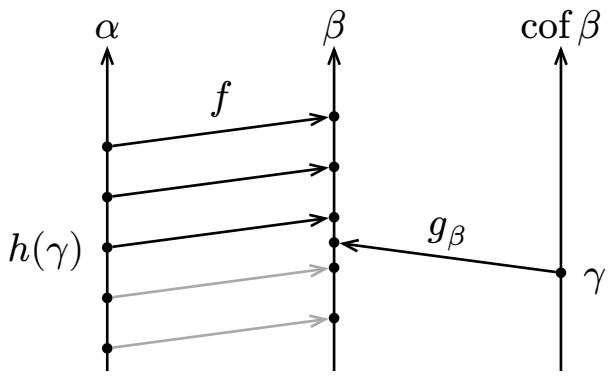
Lemma 41B. If α is a limit ordinal, and $f : \alpha \to \beta$ is a strictly increasing cofinal map, then $\operatorname{cof} \alpha = \operatorname{cof} \beta$.

Proof. First, apply Lemma 41A to get strictly increasing cofinal maps $g_\alpha : \operatorname{cof} \alpha \to \alpha$ and $g_\beta : \operatorname{cof} \beta \to \beta$. We proceed to show $\operatorname{cof} \alpha \leqslant \operatorname{cof} \beta$ and vice versa.

To show $\operatorname{cof} \beta \leqslant \operatorname{cof} \alpha$, notice that $f \circ g_\alpha : \operatorname{cof} \alpha \to \beta$ is a cofinal map. So $\operatorname{cof} \beta \leqslant \operatorname{cof} \alpha$ by definition of $\operatorname{cof} \beta$.

For the other direction, we construct a cofinal $h : \operatorname{cof} \beta \to \alpha$ as follows:

$$h(\gamma) = \min\{\xi \in \alpha : f(\xi) > g_\beta(\gamma)\}$$



We need to first verify that the set of which we are taking the minimum is nonempty. Indeed, if $\gamma \in \operatorname{cof}\beta$, then $g_\beta(\gamma) \in \beta$, so there is some $\xi' \in \alpha$ such that $f(\xi') \geqslant g_\beta(\gamma)$. Since α is a limit ordinal and f is strictly increasing, we have that $\xi' + 1 \in \alpha$ and $f(\xi' + 1) > f(\xi') \geqslant g_\beta(\gamma)$ as wanted.

To show that h maps cofinally, consider any $\xi' \in \alpha$. Then $f(\xi') \in \beta$, so there is some $\gamma \in \operatorname{cof}\beta$ such that $f(\xi') \leqslant g_\beta(\gamma)$. This establishes that $\xi' \leqslant h(\gamma)$ because if $\xi \in \alpha$ and $f(\xi) > g_\beta(\gamma)$, then $f(\xi') \leqslant g_\beta(y) < f(\xi)$, so $\xi' < \xi$ because f is strictly increasing. □

Lemma 41C. If α is a limit ordinal, then $\operatorname{cof}\omega_\alpha = \operatorname{cof}\alpha$.

Proof. Define $f : \alpha \to \omega_\alpha$ by $f(\xi) = \omega_\xi$. Notice f is strictly increasing by Lemma 32A (3). Thus we will apply Lemma 41B by showing f is a cofinal map.

Since α is a limit ordinal, $\omega_\alpha = \sup_{\xi<\alpha} \omega_\xi$ by definition, and ω_α is a limit cardinal by Lemma 32A (4). So for any $\gamma \in \omega_\alpha$, there exists an cardinal $\omega_\xi \in \omega_\alpha$ such that $\gamma \leqslant \omega_\xi = f(\xi)$. □

Remark 41A. (Group X) These statements are consistent with ZF:

- The cofinality of ω_1 is ω.
- $P(\omega)$ is a countable union of countable sets.
- ω_1 is a countable union of countable sets.

4.2 Regular Ordinals

Definition 42A. An ordinal α is **regular** iff it is a limit ordinal and $\operatorname{cof}\alpha = \alpha$.

- Similarly, a cardinal κ is **regular** iff $\operatorname{cof}\kappa = \kappa$.

Lemma 42A. If α is a regular ordinal, then α is a cardinal.

Proof. Exercise 4.1. □

Lemma 42B. If κ is an infinite cardinal, then κ^+ is a regular ordinal.

Proof. Let $g : \operatorname{cof}\kappa^+ \to \kappa^+$ be a cofinal map as per Lemma 41A. Since g is cofinal, we have that

$$\bigcup_{\alpha < \operatorname{cof}\kappa^+} g[\alpha] = \kappa^+. \tag{1}$$

Notice that $|g[\alpha]| \leqslant |\alpha| \leqslant \alpha < \operatorname{cof}\kappa^+$. Thus, if $\operatorname{cof}\kappa^+ < \kappa^+$, then $|g[\alpha]| \leqslant \kappa$, so by Lemma 32B,

$$\left| \bigcup_{\alpha < \operatorname{cof}\kappa^+} g[\alpha] \right| \leqslant \kappa \tag{2}$$

which contradicts Equation 1. □

Definition 42B. A **weakly inaccessible** cardinal is a regular limit cardinal.

Definition 42C. A **stronly inaccessible** cardinal is an cardinal $\kappa > \omega$ such that

$$\forall \alpha < \kappa \quad 2^\alpha < \kappa.$$

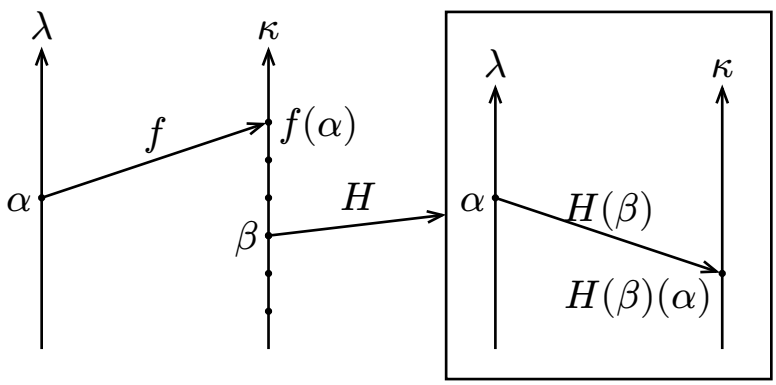
Lemma 42C. (AC) (König's) For any infinite cardinal κ and λ,

$$\operatorname{cof}\kappa \leqslant \lambda \implies \kappa^\lambda > \kappa.$$

Proof. Since $\operatorname{cof}\kappa \leqslant \lambda$, there exists a cofinal map $f : \lambda \to \kappa$. Now BWOC assume a surjection $H : \kappa \twoheadrightarrow \kappa^\lambda$ exists. We will construct a function $h : \lambda \to \kappa$ that is not in the range of H.

For $\alpha \in \lambda$, let $H_\alpha = \{H(\beta)(\alpha) : \beta < f(\alpha)\}$, we then define

$$h(\alpha) = \min(\kappa - H_\alpha). \tag{3}$$



To see why $h(\alpha)$ is well defined, it suffices to show that H_α has less elements than κ. Indeed, since H_α can be indexed by the set $f(\alpha)$, we have that $|H_\alpha| \leqslant |f(\alpha)| \leqslant f(\alpha) < \kappa$. Notice that by definition of h, we also have that

$$\forall \alpha \in \lambda \quad \forall \beta \in f(\alpha) \quad H(\beta)(\alpha) \neq h(\alpha). \tag{4}$$

To show h is not in the range of H, consider any $H(\beta) \in \operatorname{ran} H$ for $\beta \in \kappa$. Since $f : \lambda \to \kappa$ is a cofinal map and κ is a limit ordinal, there exists some $\alpha \in \lambda$ such that $\beta < \beta + 1 \leqslant f(\alpha)$. Thus by Equation 4, $H(\beta)(\alpha) \neq h(\alpha)$ and therefore $H(\beta) \neq h$. □

4.3 Continuum Hypothesis

Definition 43A. The **generalised continuum hypothesis** GCH says that:

$$2^{\omega_\alpha} = \omega_{\alpha+1} \text{ for any ordinal } \alpha.$$

The **continuum hypothesis** CH says that:

$$2^{\omega} = \omega_1$$

Lemma 43A. (AC + GCH) Let cardinals κ and λ where at least one of them is infinite.

(I) If $\lambda < \operatorname{cof} \kappa$, then $\kappa^\lambda = \kappa$

(II) If $\operatorname{cof} \kappa \leqslant \lambda < \kappa$, then $\kappa^\lambda = \kappa^+$.

(III) If $\kappa \leqslant \lambda$, then $\kappa^\lambda = \lambda^+$.

Proof. (III) Apply Lemma 34A with GCH, we have that

$$\kappa^\lambda = 2^\lambda = \lambda^+$$

□

Proof. (II) By König's Lemma, we have that $\kappa < \kappa^\lambda$, thus $\kappa^+ \leqslant \kappa^\lambda$ by definition of successor cardinal. The other direction follows from Lemma 34A with GCH. In particular,

$$\kappa^\lambda \leqslant \kappa^\kappa = 2^\kappa = \kappa^+.$$

□

Proof. (I) An injection from κ to κ^λ can be constructed by constant functions. Thus it is left to show $\kappa^\lambda \leqslant \kappa$.

First notice that $\lambda < \operatorname{cof} \kappa$ means there is no cofinal map from λ into κ. In other words, the range of any function $f : \lambda \to \kappa$ is bounded in the limit ordinal κ. Therefore, $f : \lambda \to \kappa$ iff $f : \lambda \to \alpha$ for some $\alpha < \kappa$, so

$${}^\lambda\kappa = \bigcup_{\alpha<\kappa} {}^\lambda\alpha.$$

Notice that GCH implies that $|{}^\lambda\alpha| \stackrel{\text{def}}{=} \alpha^\lambda \leqslant \max\{\alpha^+, \lambda^+\} \leqslant \kappa$, so by Lemma 32B,

$$\kappa^\lambda = \left| \bigcup_{\alpha<\kappa} {}^\lambda\alpha \right| \leqslant \kappa.$$

□

Exercises

Exercise 4.1. (Group Y) Prove Lemma 42A.

5 Combinatorial Set Theory

5.1 Almost Disjoint Families

Definition 51A. Given cardinal $\kappa \geqslant \omega$, an **almost disjoint (AD) family** on κ is a family $\mathcal{A} \subseteq \mathcal{P}(\kappa)$ such that for any distinct $X, Y \in \mathcal{A}$

$$|X| = \kappa \text{ and } |X \cap Y| < \kappa.$$

If no proper superset of $\mathcal{A}$ is AD, we say $\mathcal{A}$ is a **maximal almost disjoint (MAD)** family.

Almost disjoint families are used for combinatorial set theory, which was the set theory used before the introduction of independence proofs.

Remark 51A. There is no pairwise disjoint family on κ of cardinality larger than κ. It follows from pigeonhole principle.

Theorem 51A. Let $\kappa \geqslant \omega$ be an regular cardinal.
(I) No almost disjoint family on κ with cardinality κ is maximal.
(II) There exists a MAD family on κ with cardinality larger than κ.

Proof. (I) Let any almost disjoint family $\mathcal{A} = \{A_\alpha \in [\kappa]^\kappa : \alpha < \kappa\}$, we use diagonalisation to show there exists a set $B \in [\kappa]^\kappa$ such that

$$B \notin \mathcal{A} \text{ and } \forall \alpha < \kappa \ \ |A_\alpha \cap B| < \kappa.$$

To begin, for any $\alpha < \kappa$, recursively define

$$B_\alpha = A_\alpha - \bigcup_{\xi<\alpha} (A_\alpha \cap A_\xi).$$

Since κ is a regular cardinal, the set difference is nonempty (why?). Thus, pick $b_\alpha \in B_\alpha$ for any $\alpha < \kappa$, we argue that $B = \{b_\alpha : \alpha < \kappa\}$ is our desired set.

Intuitively, B_α is defined as the subset of A_α in which we remove A_ξ for any $\xi < \alpha$. Thus the b_α's are unique, so $|B| = \kappa$. Furthermore, for any $\alpha < \kappa$, we have that

$$A_\alpha \cap B \subseteq \{b_\xi : \xi \leqslant \alpha\} \in [\kappa]^{<\kappa}.$$

Finally, $B \notin \mathcal{A}$ because for any $\alpha < \kappa$, we have that $b_{\alpha+1} \notin A_\alpha$. □

Corollary 51B. Assuming GCH, if $\mathcal{A} \subseteq \mathcal{P}(\kappa)$ is MAD then $|\mathcal{A}| = \kappa^+$.

Theorem 51C. For any infinite cardinal κ such that $2^{<\kappa} = \kappa$, there exists an almost disjoint family on κ with cardinality 2^κ.

Proof. We first consturct a set I with cardinality κ and a family $\mathcal{A}' \subseteq \mathcal{P}(I)$ with cardinality 2^κ and almost disjoint on κ. Then, we use the bijection from I to κ to translate $\mathcal{A}'$ to our desired almost disjoint family on κ.

Let $I = \{F \subseteq \kappa : \sup F < \kappa\}$ (why not just $|F| < \kappa$?). Observe that $I \subseteq [\kappa]^{<\kappa}$ and $|I| = \kappa$ (why?). Next, we construct a family on I; for any $x \in [\kappa]^\kappa$, define

$$A_x = \{x \cap \alpha : \alpha < \kappa\} \subseteq I.$$

Let $\mathcal{A}' = \{A_x : x \in [\kappa]^\kappa\} \subseteq \mathcal{P}(I)$, we have that for any distinct A_x and $A_y \in \mathcal{A}'$,

$$|A_x| = \kappa \text{ and } |A_x \cap A_y| < \kappa.$$

To see why $|A_x \cap A_y| < \kappa$, WLOG assume there exists $\eta \in y - x$. Then $x \cap \alpha \neq y \cap \alpha$ for any $\alpha \geqslant \eta$ because $\eta \in y \cap \alpha$ whereas $\eta \notin x \cap \alpha$. Since A_x and A_y has no common elements beyond the index η, we establish that $|A_x \cap A_y| \leqslant \eta < \kappa$ as desired.

Now we have the conditions of being almost disjoint, what is left is to translate each A_x into a subset of κ. To this end, let $\varphi : I \xrightarrow{\sim} \kappa$ be a bijection, define

$$\mathcal{A} = \{\varphi[A_x] : x \in [\kappa]^\kappa\} \subseteq \mathcal{P}(\kappa).$$

Since φ is injective, we have that $|\varphi[A_x]| = |A_x| = \kappa$, and for any distinct $\varphi[A_x]$ and $\varphi[A_y]$,

$$|\varphi[A_x] \cap \varphi[A_y]| = |\varphi[A_x \cap A_y]| = |A_x \cap A_y| < \kappa.$$

Finally, $\mathcal{A}$ has the same cardinality as $\mathcal{A}'$, which is $\kappa^\kappa = 2^\kappa$. □

5.2 Delta-Systems

Definition 52A. For family $\mathcal{F}$, a set r is the **root** of $\mathcal{F}$ iff for any distinct $X, Y \in \mathcal{F}$,

$$X \cap Y = r.$$

We say $\mathcal{F}$ forms a **Δ-system** iff there exists such root.

Theorem 52A. Let κ be an infinite cardinal and $\theta > \kappa$ be a regular cardinal such that

$$\forall \alpha < \theta \quad \alpha^{<\kappa} < \theta.$$

If A is a family of cardinality at least θ, and

$$\forall x \in A \quad |x| < \kappa,$$

then there exists a Δ-system $B \subseteq A$ such that $|B| = \theta$.

For the proof Theorem 52A, there are two important observations regarding a set X having cardinality θ (a regular cardinal).

Remark 52A. A subset $A \subseteq X$ is unbounded (i.e. cofinal) if and only if $|A| = \theta$.

Remark 52B. X is not the less than θ union of subsets with cardinality less than θ. In other words, if X is the less than θ union of some subsets, some of them must have cardinality θ.

Proof.

First, we will shrink A until $|A| = \theta$. Thus $|\bigcup A| \leqslant \theta$ because $\bigcup A$ is now a θ union of sets with cardinality less than θ. So WLOG assume $\bigcup A \subseteq \theta$, i.e. $A \subseteq [\theta]^{<\kappa}$.

Since each $x \in A$ has cardinality strictly less than κ, so is its order type. Therefore we can rewrite A as a κ $(< \theta)$ union of subsets:

$$A = \bigcup_{\eta<\kappa} \{x \in A : \operatorname{otp} x = \eta\}.$$

Because $|A| = \theta$ where θ is a regular cardinal, by Remark 52B, there is some $\eta < \kappa$ such that $A_1 = \{x \in A : \operatorname{otp} x = \eta\}$ has cardinality θ. Fix such $\eta < \kappa$, now we will only consider the elements in $A_1 \subseteq A$.

Now A_1 consists of θ elements whose cardinality is less than κ. However, for each $\alpha < \theta$, there are only $\alpha^{<\kappa} < \theta$ subsets of α whose cardinality is also less than κ. So there is some $x \in A_1$ such that $x \nsubseteq \alpha$, meaning $\alpha \leqslant \delta$ for some ordinal $\delta \in x \subseteq \bigcup A_1$. This establishes that $\bigcup A_1$ is a cofinal subset of θ, so $|\bigcup A_1| = \theta$ by Remark 52A.

Since each $x \in A_2$ has exactly order type η, for $\xi < \eta$, let $x(\xi) \in \theta$ denote the ξth element of x (as per $\operatorname{otp} x$). Thus, we can again write $\bigcup A_1$ as a η $(< \theta)$ union:

$$\bigcup A_1 = \bigcup_{\xi<\eta} \{x(\xi) : x \in A_1\}.$$

Since $\bigcup A_1$ has cardinality θ (a regular cardinal), there is some $\xi < \eta$ such that $\{x(\xi) : x \in A_1\}$ has cardinalily θ (Remark 52A). Let $\xi_0 < \eta$ be the least of such ξ.

Next, define

$$\alpha_0 = \sup_{\substack{x \in A_1 \\ \rho < \xi_0}} x(\rho) + 1.$$

It follows from definition that $x(\rho) < \alpha_0$ for each $x \in A_1$ and $\rho < \xi_0$. We also note that $\alpha_0 < \theta$. This is because ξ_0 was chosen to be the least ordinal such that $\{x(\xi_0) \in \theta : x \in A_1\}$ is unbounded in θ. So for $\rho < \xi_0$, the set $\{x(\rho) \in \theta : x \in A_1\}$ is bounded. Therefore $\bigcup_{x \in A_1} x(\rho) + 1 < \theta$, so $\alpha_0 = \bigcup_{\rho < \xi_0} \bigcup_{x \in A_1} x(\rho) + 1$ is a ξ_0 ($< \theta$) union of sets whose cardinality is less than θ. By Remark 52B, this implies $\alpha_0 < \theta$.

Using α_0, we proceed to shrink A_1 further. By recursion on $\mu < \theta$, we will choose $x_\mu \in A_1$ such that

$$x_\mu(\xi_0) > \alpha_0 \text{ and } x_\mu(\xi_0) > \sup_{\substack{\nu < \mu \\ \rho < \eta}} x_\nu(\rho).$$

Notice that such $x_\mu \in A_1$ always exists because $\{x(\xi_0) : x \in A_1\}$ has cardinality θ, whereas α_0 and $\sup\{x_\nu(\rho) : \nu < \mu \wedge \rho < \eta\}$ are both less than θ (the latter is again a μ ($< \theta$) union of sets with cardinality less than θ).

Since all the x_μ's are distinct (they disagree at the ξ_0th element), define $A_2 = \{x_\mu \in A_1 : \mu < \theta\}$, we then have that $|A_2| = \theta$ and for any distinct $x_\mu, x_\nu \in A_2$,

$$x_\mu \cap x_\nu \subseteq \alpha_0$$

Indeed, if $\rho < \xi_0$, then $x_\mu(\rho) \in \alpha_0$ for each $x_\mu \in A_2$. Otherwise, if $\rho \geqslant \xi_0$ and WLOG $\nu < \mu$, then $x_\mu(\rho) \notin x_\nu$ because $x_\mu(\rho) \geqslant x_\mu(\xi_0) > x_\nu(\rho')$ for any $\rho' < \eta$ by our choice of x_μ.

Next, we will rewrite $A_2 \in [A_1]^\theta$ as a $\alpha_0^{<\kappa}$ union of subsets:

$$A_2 = \bigcup_{r \in [\alpha_0]^{<\kappa}} \{x_\mu \in A_2 : x_\mu \cap \alpha_0 = r\}.$$

Since $\alpha_0 < \theta$, we have that $\alpha_0^{<\kappa} < \theta$, meaning that A_2 is a $< \theta$ union of subsets. Thus by Remark 52B, there is some $r \in [\alpha_0]^{<\kappa}$ such that $B = \{x_\mu \in A_2 : x_\mu \cap \alpha_0 = r\}$ has cardinality θ.

Finally, $B \subseteq A$ forms a Δ-system with root r because for each distinct $x_\mu, x_\nu \in B$, we have that $x_\mu \cap x_\nu \subseteq \alpha_0$, so

$$x_\mu \cap x_\nu = (x_\mu \cap \alpha_0) \cap (x_\nu \cap \alpha_0) = r.$$

□

Here we present an application of this theorem in topology.

Example 52A. (Topology) If $\{X_\alpha : \alpha \in \Lambda\}$ is a collection of ccc topological spaces such that for each finite subset $F \in [\Lambda]^{<\omega}$, the product $\prod_{\alpha \in F} X_\alpha$ is ccc, then

$$X = \prod_{\alpha \in \Lambda} X_\alpha$$

is ccc.

Proof. BWOC assume X is not ccc. WLOG we can assume that there exists a collection

$$\{U_\alpha \subseteq X : \alpha < \omega_1\}$$

consisting of nonempty pairwise disjoint basic open sets. Each U_α being basic means that it is a basis element in the product topology. So, we can identify each U_α by some $F_\alpha \in [\Lambda]^{<\omega}$.

Now we are ready to apply Theorem 52A. In particular, the family of interest is

$$A = \{F_\alpha : \alpha < \omega_1\}.$$

Let $B \in [A]^{\omega_1}$ be a Δ-system with root $r \in [\Lambda]^{<\omega}$. Notice that r cannot be empty; otherwise, if $F_\alpha \cap F_\beta = \varnothing$ for $\alpha, \beta \in r$, then U_α and U_β are not disjoint since we can pick an $x \in X$ whose F_α coordinates are in U_α and F_β coordinates in U_β, thus making $x \in U_\alpha \cap U_\beta$.

Finally, consider the projection

$$\pi_r : \prod_{\alpha \in \Lambda} X_\alpha \to \prod_{i \in r} X_i.$$

Since $\{U_\alpha : F_\alpha \in B\}$ is pairwise disjoint, so is the projection $\{\pi_r(U_\alpha) : \alpha \in B\}$, but the latter is an uncountable collection of nonempty open sets, contradicting that $\prod_{i \in r} X_i$ is ccc. □

Example 52B. Every separable space is ccc, but the converse does not hold in general. For instance, for $\kappa > 2^\omega$, the product space 2^κ (where 2 is the discrete space $\{0, 1\}$) is ccc, but not separable.

6 Martin's Axioms

Definition 6A. A statement φ is **independent** from ZFC iff ZFC $\nvdash \varphi$ and ZFC $\nvdash \neg\varphi$.

- i.e. ZFC cannot prove or disprove φ.
- i.e. φ and $\neg\varphi$ are both **consistent** with ZFC.
- i.e. ZFC, φ leads to no contradictions.
- i.e. there is no statement γ such that ZFC, $\varphi \vdash \gamma \wedge \neg\gamma$.

Remark 6A. There are statements consistent with ZFC, but their dependency is not settled.

6.1 Posets and CCC

Definition 61A. A **poset** is a set P endowed with a partial order $\leqslant$. That is, it satisfies:

- *reflexivity*: $\forall a \in P \quad a \leqslant a$
- *transitivity*: $\forall a, b, c \in P \quad a \leqslant b \wedge b \leqslant c \Longrightarrow a \leqslant c$

In forcing, elements $p, q \in P$ are called "forcing conditions": $p \leqslant q$ means p *extends* q (i.e. p is stronger than q).

Given a poset P, we have the following definitions:

Definition 61B. $C \subseteq P$ is a **chain** iff

$$\forall p, q \in C \quad p \leqslant q \vee q \leqslant p$$

Definition 61C. $p, q \in P$ are **compatible** iff

$$\exists r \in P \quad r \leqslant p \wedge r \leqslant q$$

- i.e. p and q have some **common extension** $r \in P$.

Otherwise p and q are **incompatible**, denoted by $p \perp q$.

Definition 61D. $p, q \in P$ are **comparable** iff

$$\exists s \in P \quad p \leqslant s \wedge q \leqslant s$$

Definition 61E. $D \subseteq P$ is **dense** iff

$$\forall p \in P \quad \exists d \in D \quad d \leqslant p$$

Definition 61F. $A \subseteq P$ is an **antichain** iff

$$\forall p, q \in A \quad p \neq q \Longrightarrow p \perp q$$

Definition 61G. $G \subseteq P$ is a **filter** iff

1. every pair of elements in G is compatible, i.e.

$$\forall p, q \in G \quad \exists r \in G \quad r \leqslant p \wedge r \leqslant q.$$

2. G is *closed upwards*, i.e.

$$\forall p \in G \quad \forall q \in P \quad p \leqslant q \Longrightarrow q \in G.$$

Definition 61H. A poset P has the **countable chain condition** (ccc) iff every antichain is countable.

Example 61A. Consider ω_1 under the usual ordering. Notice that if $A \subseteq \omega_1$ is an antichain, then $|A| \leqslant 1$ since every pair of ordinals is compatible.

6.2 Martin's Axioms (Set)

Definition 62A. For an cardinal $\omega \leqslant \kappa < 2^\omega$, **Martin's axiom** (set) $\mathsf{MA}(\kappa)$ says that:

For any ccc poset P and any family of dense subsets $\mathcal{D} = \{D_\alpha : \alpha \leqslant \kappa\}$, there exists a filter $G \subseteq P$ such that $\forall \alpha \leqslant \kappa \quad G \cap D_\alpha \neq \varnothing$.

In general, MA says that $\mathsf{MA}(\kappa)$ holds for any cardinal $\omega \leqslant \kappa < 2^\omega$.

Theorem 62A. ZFC implies $\mathsf{MA}(\omega)$.

Proof. Consider any ccc poset P and any family of dense subsets $\mathcal{D} = \{D_n \subseteq P : n \leqslant \omega\}$. Since P is nonempty, pick some $x \in P$, we then have some $d_0 \in D_0$ such that $d_0 \leqslant x$, since D_0 is dense.

Now, for any $n \in \omega$, we can pick some $d_{n+1} \in D_{n+1}$ such that $d_{n+1} \leqslant d_n$. By AC, collect them into a set $\{d_n : n \in \omega\}$. Let $G = \{p \in P : \exists n (d_n \leqslant p)\}$, we have that G is a filter that intersects each D_n at d_n. □

G is a filter because for any $p, q \in G$, there is some $d_n \leqslant p$ and $d_m \leqslant q$. So p and q are compatible by $d_{\min\{n,m\}}$ (transitivity). Similarly, for any $p \in G$ and any $q \geqslant p$, there is some $d_n \leqslant p$, so $d_n \leqslant q$ as well.

Theorem 62B. ZFC implies $\neg\, \mathsf{MA}(2^\omega)$.

Proof.

Consider the poset $(\mathrm{Fn}(\omega, 2), \leqslant)$ defined as

$$\mathrm{Fn}(\omega, 2) \stackrel{\text{def}}{=} \{p : p \text{ is a function}, \operatorname{dom} p < \omega, \operatorname{ran} p \subseteq 2\}$$
$$p \leqslant q \Longleftrightarrow p \supseteq q$$

- Intuitively, $\mathrm{Fn}(\omega, 2)$ is the set of finite partial functions from ω into $\{0, 1\}$.

First note that $\mathrm{Fn}(\omega, 2)$ has ccc since $|\mathrm{Fn}(\omega, 2)| = \aleph_0$, so every antichain is countable. Notice $p \perp q$ iff p and q don't agree on their common domain.

Now, for any $f : \omega \to 2$ and any $n \in \omega$, define these dense sets:

$$D_n = \{p \in \mathrm{Fn}(\omega, 2) : n \in \operatorname{dom} p\}$$
$$D_f = \{p \in \mathrm{Fn}(\omega, 2) : \exists n(p(n) \neq f(n))\}$$

Let $\mathcal{D} = \{D_\alpha : \alpha \leqslant 2^\omega\}$ be the family of dense sets as defined above. To see why they are dense, consider any $p \in \mathrm{Fn}(\omega, 2)$:

- For any $D_n \in \mathcal{D}$, we have that $p \in D_n$ or $p \cup \{(n, 0)\} \in D_n$, which are both extensions of p.
- Consider any $D_f \in \mathcal{D}$. Since $\operatorname{dom} p$ is finite, there exists some $n \in \omega - \operatorname{dom} p$. WLOG assume $f(n) = 1$, thus $p \cup \{(n, 0)\} \leqslant p$, which is in D_f.

BWOC assume there is a filter $G \subseteq \mathrm{Fn}(\omega, 2)$ that intersects each D_f and each D_n. Let $g = \bigcup G \subseteq \omega \times 2$, we will show that $g : \omega \to 2$.

Indeed, since $G \cap D_n \neq \varnothing$ for any $n \in \omega$, we have that $\operatorname{dom} g = \omega$. If $(n, a), (n, a') \in g$, then for some $p, p' \in G$, we have that $p(n) = a$ and $p'(n) = a'$. Since p and p' are compatible (being elements of a filter), they must agree on their common domain: in particular, $a = a'$.

Therefore $g : \omega \to 2$, so G must intersect D_g at some $p \in \mathrm{Fn}(\omega, 2)$. However, this leads to the conclusion that $p \subseteq g$ but $\exists n(p(n) \neq g(n))$, a contradiction. □

Example 62A. (Group X) Consider poset $P = \mathrm{Fn}(\omega, \omega_1)$ with

$$p \leqslant q \iff p \supseteq q$$

Define the dense subsets as follows:

$$\forall n \in \omega \quad D_n = \{p \in P : n \in \operatorname{dom} p\}$$
$$\forall \alpha \in \omega_1 \quad D_\alpha = \{p \in P : \exists n_\alpha \in \omega \quad p(n_\alpha) = \alpha\}$$

BWOC assume there is a filter $G \subseteq P$ that intersects at each dense set.

- Since G is a filter, we have that $g = \bigcup G$ is a function.
- Since G intersects each D_n, we have that $g : \omega \to \omega_1$.
- Since G intersects each D_α, we have that $g : \omega \twoheadrightarrow \omega_1$, contradicts $|\omega| < |\omega_1|$.

Notice that $(P, \leqslant)$ doesn't have ccc: the antichain $\{(0, \alpha) : \alpha \in \omega_1\}$ is not countable.

6.3 Topologies and CCC

Definition 63A. A topological space has **ccc** (countable chain condition) iff every family of nonempty pairwise disjoint open sets is countable.

- e.g. $(\mathbb{R}, \mathcal{T}_E)$ has ccc because $\mathbb{Q}$ is dense in $\mathbb{R}$

Example 63A. Let any ccc topological space $(X, \mathcal{T})$, define $P_X = \{p \in \mathcal{T} : p \neq \varnothing\}$ with partial order $p \leqslant q$ iff $p \subseteq q$. Notice $p \perp q$ iff $p \cap q = \varnothing$, thus the space $(X, \mathcal{T})$ has ccc iff the poset $(P_X, \leqslant)$ has ccc.

For a topological space $(X, \mathcal{T})$, we have the following definitions:

Definition 63B. $D \subseteq X$ is **dense** iff D intersects every open set in X.

Definition 63C. $N \subseteq X$ is **nowhere dense** (nwd) iff $\left(\overline{N}\right)^\circ = \varnothing$

- In other words, for any nonempty open set U, there is a nonempty open subset $V \subseteq U$ disjoint from N.

Example 63B. $\mathbb{Z}$ is nowhere dense in $\mathbb{R}$, and a line is nowhere dense in $\mathbb{R} \times \mathbb{R}$.

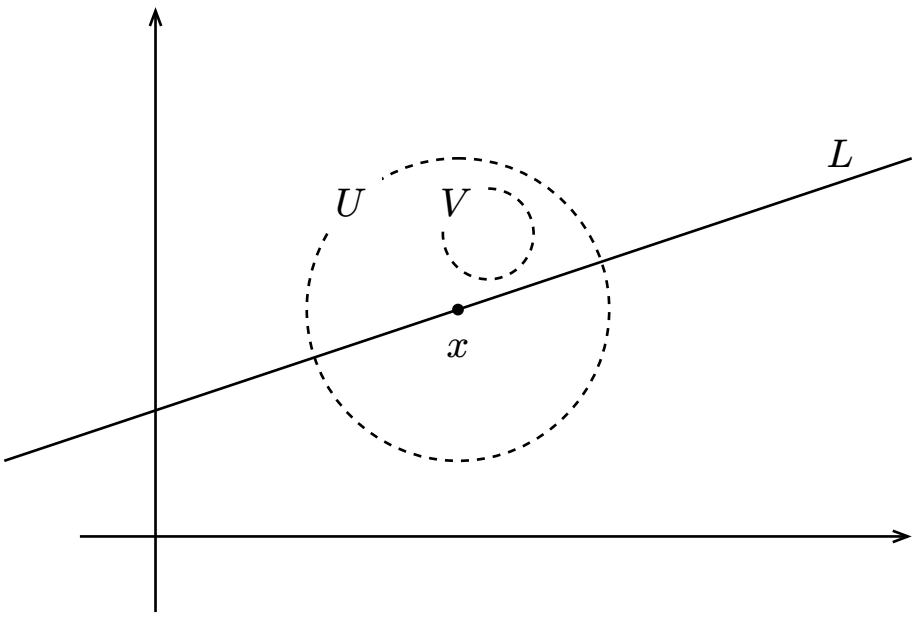


Figure 9: A line L in $\mathbb{R}^2$ is nowhere dense.

Definition 63D. $A \subseteq X$ is **meager** iff A is a countable union of nowhere dense sets.

- A meager set is also called a set **of first category**.

Definition 63E. Topological space X is **compact** iff any family of closed sets $\mathcal{K}$ satisfying *FIP* has a nonempty intersection.

- $\mathcal{K}$ satisfies the *finite intersection property* (FIP) iff $\forall \mathcal{K}_0 \in [\mathcal{K}]^{<\aleph_0} \quad \bigcap \mathcal{K}_0 \neq \varnothing$

Definition 63F. Topological space X is (Hausdorff) **regular** iff for any closed $K \subseteq X$ and $x \notin K$, there exists $U \in \eta(x)$ and open $V \supseteq K$ such that $U \cap V = \varnothing$.

- That is, the closed set K and the point x can be separated by open neighborhoods.

Theorem 63A. Topological space X is regular iff

$$\forall x \in X \quad \forall U \in \eta(x) \quad \exists V \in \eta(x) \quad \overline{V} \subseteq U$$

- That is, every open neighborhood contains the closure of a smaller open neighborhood.

Proof.

($\Longrightarrow$) Consider any $x \in X$ and $U \in \eta(x)$
$\therefore X - U$ is closed and $x \notin X - U$.
$\therefore \exists V \in \eta(x) \quad \exists W \supseteq X - U$ such that $V \cap W = \varnothing$.
$\therefore V \subseteq X - W$ where $X - W$ is closed.
$\therefore \overline{V} \subseteq X - W \subseteq U$.

($\Longleftarrow$) Consider any closed $K \subseteq X$ and any $x \in X - K$.
$\therefore X - K \in \eta(x)$, thus $\exists V \in \eta(x)$ such that $\overline{V} \subseteq X - K$.
$\therefore X - \overline{V} \subseteq K$, which is disjoint with V, since $X - \overline{V} \subseteq X - V$. □

6.4 Martin's Axioms (Topology)

Definition 64A. For an cardinal $\omega \leqslant \kappa < 2^{\omega}$, **Martin's axiom** (topology) $\mathsf{MA}(\kappa)$ says that:

No ccc compact Hausdorff space is the union of κ many closed nwd sets.

In other words:

For any ccc compact T_2 space X, and dense open $\{D_\alpha \subseteq X : \alpha \leqslant \kappa\}$,

$$\bigcap_{\alpha \leqslant \kappa} D_\alpha \neq \varnothing.$$

Notice that a compact T_2 space is regular.

Theorem 64A. Set $\mathsf{MA}(\kappa)$ iff topology $\mathsf{MA}(\kappa)$

Proof.

($\Longrightarrow$) Consider any ccc compact T_2 space $(X, \mathcal{T})$ and any dense open $\{U_\alpha : \alpha \leqslant \kappa\}$.

First, let $(P, \leqslant)$ be the ccc poset as defined in Example 63A. That is,

$$P = \{p \in \mathcal{T} : p \neq \varnothing\}$$
$$p \leqslant q \Longleftrightarrow p \subseteq q$$

Next, we construct the dense set $D_\alpha \subseteq P$ for an $\alpha \leqslant \kappa$:

$$D_\alpha = \left\{d \in P : \overline{d} \subseteq U_\alpha\right\}$$

Each D_α are dense because for any $p \in P$, pick some $x \in p \cap U_\alpha$ as U_α is dense. Then $p \cap U_\alpha \in \eta(x)$, thus there is some $V \in \eta(x)$ such that $\overline{V} \subseteq p \cap U_\alpha \subseteq U_\alpha$ because X is regular. Finally, $V \in D_\alpha$ and $V \leqslant p$ because $V \subseteq \overline{V} \subseteq p$.

With this setup, we can apply set $\mathsf{MA}(\kappa)$ to get a filter $G \subseteq P$ and chooose $p_\alpha \in G \cap D_\alpha$ for any $\alpha \leqslant \kappa$. Defne $\mathcal{K} = \{\overline{p_\alpha} \subseteq X : a \leqslant \kappa\}$ to be a family of closed subsets.

The family $\mathcal{K}$ has FIP, since for any $\overline{p_1}, \cdots, \overline{p_n} \in \mathcal{K}$, there exists $r \in G$ such that $r \leqslant p_i$ for any $1 \leqslant i \leqslant n$. Thus $\varnothing \neq r \subseteq \bigcap_i p_i \subseteq \bigcap_i \overline{p_i}$. Therefore, we have $\varnothing \neq \bigcap \mathcal{K} \subseteq \bigcap_{\alpha \leqslant \kappa} D_\alpha$ as wanted.

($\Longleftarrow$) see Exercise 6.3 $\square$

6.5 Applications of Martin Axioms

Let κ be an infinite cardinal, where $\omega \leqslant \kappa < 2^\omega$. Consider the following questions:

Example 65A. Is $2^\kappa = 2^\omega$?

Example 65B. If $\mathcal{A}$ is an almost disjoint family of size κ, does $\mathcal{A}$ fail to be MAD (maximal almost disjoint)?
- Notice for $\kappa = \omega$, the statement holds

Example 65C. (measure theory) Does the union of κ many subsets of $\mathbb{R}$ of which eash has Lebesgue measure 0 have Lebesgue measure 0?

Martin's axiom (MA) allows us to answer these questions. In particular, MA implies that Example 65A, Example 65B, and Example 65C all hold. In other words, for $\omega \leqslant \kappa < 2^\omega$, we have that κ behaves similar to ω.

Theorem 65A. (Solovay) Assume $\mathsf{MA}(\kappa)$, let $A, C \in [\mathcal{P}(\omega)]^\kappa$ satisfying

$$\forall y \in C \quad \forall F \in [A]^{<\omega} \quad |y - \bigcup F| = \omega,$$

then there exists $d \in [\omega]^\omega$ such that
1. $\forall x \in A \quad |x \cap d| < \omega$,
2. $\forall y \in C \quad |y \cap d| = \omega$.

Proof. Define

$$P_A = \left\{\langle s, F\rangle : s \in [\omega]^\omega \wedge F \in [A]^{<\omega}\right\}$$
$$\langle s', F'\rangle \leqslant \langle s, F\rangle \Longleftrightarrow s' \supseteq s \wedge F' \supseteq F \wedge \forall x \in F \quad x \cap s' \subseteq s$$

Notice if G is a generic, then $\langle s, F\rangle$ forces $\forall x \in F \quad \forall n \notin x - s \quad n \notin d$, and $\langle s', F'\rangle \leqslant \langle s, F\rangle$ forces $\forall x \in F \quad \forall n \in x - s \quad n \notin s'$, meaning we can't put n in s' that was forces to not be in d.

First, remark that $\langle s_1, F_1\rangle \perp \langle s_2, F_2\rangle$ iff

$$\forall x \in F_1 x \cap s_2 \subseteq s_1 \wedge \forall x \in F_2 \quad x \cap s_1 \subseteq s_2. \tag{5}$$

In which case $\langle s_1 \cup s_2, F_1 \cup F_2\rangle$ is a common extension. Notice that Equation 5 iff

$$\forall x \in F_1 \quad \forall n \in x - s_1 \quad n \notin s_2 \wedge \forall x \in F_2 \quad \forall n \notin x - s_2 \quad n \notin s_1. \tag{6}$$

Thus, P_A has ccc, for if $\langle s_\alpha, F_\alpha\rangle \perp \langle s_\beta, F_\beta\rangle$, then as elements of $[\omega]^{<\omega}$, we have that $s_\alpha \neq s_\beta$. Therefore if $\{\langle s_\alpha, F_\alpha\rangle : \alpha < \omega_1\}$ is an antichain, each $s_\alpha \in [\omega]^{<\omega}$ must be distinct. However, $[\omega]^{<\omega}$ is countable.

Now, if $G \subseteq P_A$ is a generic, we let $d_G = \bigcup\{s : \exists F \ \langle s, F\rangle \in G\}$. Thus, if $\langle s, F\rangle \in G$, then $s \subseteq d_G$ and $\forall x \in F(d_G \cap x \subseteq s)$, beacuse for each $\langle s', F'\rangle \in G$, we have that $x \cap s' \subseteq s$.

For $x \in A$, define

$$D_x = \{\langle s, F\rangle \in P_A : x \in F\}. \tag{7}$$

Note that D_x is dense because for each $\langle s, F\rangle \in P_A$, we have that $\langle s, F\rangle \geqslant \langle s, F \cup \{x\}\rangle \in D_x$

For $y \in C$ and $n \in \omega$, define

$$E_n^y = \{\langle s, F\rangle \in P_A : s \cap y \nsubseteq n\}. \tag{8}$$

Each E_n^y is also dense, because for $\langle s, F\rangle \in P_A$, pick $m \in y - \bigcup F$ such that $m > n$ (recall that $y - \bigcup F$ is infinite), then $\langle s, F\rangle \geqslant \langle s \cup \{m\}, F\rangle \in E_n^y$ because $\forall x \in F(m \notin x)$.

Finally, apply $\mathsf{MA}(\kappa)$ on $\{D_x\} \cup \{E_n^y\}$, we get an generic $G \subseteq P_A$. Pick $\langle s, F\rangle \in G \cap D_x$, we conclude that $|x \cap d_G| < \omega$ because $x \cap d \subseteq s \in [\omega]^{<\omega}$. Now, for all $y \in C$ and $n \in \omega$, pick $\langle s, F\rangle \in G \cap E_n^y$, we have that $|y \cap d_G| \nsubseteq n$ as $s \subseteq d_G$, meaning $|y \cap d| = \omega$. □

Corollary 65B. Assume $\mathsf{MA}(\kappa)$, if $A \in [\mathcal{P}(\omega)]^\kappa$ is an almost disjoint family, then A is not maximal.

Proof. Let $C = \{\omega\}$. Note that for each $F \in [A]^{<\omega}$, we have that $|\omega - \bigcup F| = \omega$. To see this, let $F = \{x_1, \cdots, x_n\}$ and pick $y \in A - F$. Thus

$$y \cap \bigcup F = y \cap (x_1 \cup \cdots \cup x_n) = (y \cap x_1) \cup \cdots \cup (y \cap x_n).$$

Thus $y \cap \bigcup F$ is a finite union of finite subsets, thus finite, meaning $y - \bigcup F$ is infinite.

Apply the theorem, we get that d_G is almost disjoint from each $x \in A$. □

Corollary 65C. Assume $\mathsf{MA}(\kappa)$, if $B \in [\mathcal{P}(\omega)]^\kappa$ is an almost disjoint family, and $A \subseteq B$. Then there exists $d \subseteq \omega$ such that

1. $\forall x \in A \quad |d \cap x| < \omega$
2. $\forall x \in B - A \quad |d \cap x| = \omega$

Proof. Apply the theorem with $C = B - A$ □

Corollary 65D. Assume $\mathsf{MA}(\kappa)$, then $2^\kappa = 2^\omega$.

Proof. Choose an almost disjoint family $B \in [\mathbb{V}]^\kappa$. Define $\varphi : \mathcal{P}(\omega) \to \mathcal{P}(B)$ by

$$\varphi(d) = \{x \in B : |d \cap x| < \omega\}.$$

Thus the previous corollary says that $\varphi : \mathcal{P}(\omega) \twoheadrightarrow \mathcal{P}(B)$. Thus

$$2^\kappa = |\mathcal{P}(B)| \leqslant |\mathcal{P}(\omega)| = 2^\omega$$

□

Corollary 65E. Assume $\mathsf{MA}(\kappa)$, then 2^ω is regular.

Proof. We show $2^\omega \leqslant \operatorname{cof} 2^\omega \leqslant 2^\omega$, thus $\operatorname{cof} 2^\omega = 2^\omega$.

Notice that for each $\kappa < 2^\omega$

$$(2^\omega)^\kappa = 2^{\omega \otimes \kappa} = (2^\kappa)^\omega = 2^{\omega \otimes \omega} = 2^\omega.$$

Thus $\kappa < \operatorname{cof} 2^\kappa$ by Konig's. □

Exercises

Exercise 6.1. (Group X) Show that subset N in space $(X, \mathcal{T})$ is nowhere dense iff

$$\forall x \in X \quad \forall U \in \eta(x) \quad \exists V \in \mathcal{T} \quad V \subseteq U \text{ and } V \cap N = \varnothing$$

Exercise 6.2. (Group X) Show that the complement of a open dense subset is closed nowhere dense.

Exercise 6.3. (Group X) Show that topology $\mathsf{MA}(\kappa)$ implies set $\mathsf{MA}(\kappa)$

Exercise 6.4. S is a *reaping family* of infinitely many subsets of ω iff

$$\forall x \in [\omega]^\omega \quad \exists s \in S \quad s \subseteq^* x \vee s \subseteq^* \omega - x.$$

That is, for each $x \in [\omega]^\omega$, there is some $s \in S$ which x or its complement almost contains.

(a) Let $\mathfrak{r}$ be the minimum cardinality of a reaping family, show that $\mathfrak{r}$ is uncountable.

(b) Assume $\mathsf{MA}(\kappa)$ only on the poset $\mathrm{Fn}(\omega, 2)$ (defined in th proof of Theorem 62B), show that $\mathfrak{r} > \kappa$.

7 Trees

Definition 7A. (Tree) A **tree** is a poset $(T, <)$ such that for each $t \in T$, the set of predecessors $\{s \in T : s < t\}$ is well-ordered by $<$.

- We say a tree is **rooted** iff T has an unique minimal element.

Definition 7B. (Height of a node) For tree T and $x \in T$, the **height** of x in T is

$$\mathrm{ht}_T(x) \stackrel{\text{def}}{=} \mathrm{otp}\{t \in T : t < x\}.$$

Definition 7C. (Level) For an ordinal α, the αth **level** of T is

$$\mathrm{Lev}_\alpha T \stackrel{\text{def}}{=} \{t \in T : \mathrm{ht}_T(t) = \alpha\}.$$

Definition 7D. (Height of a tree) The **height** of a tree T is

$$\mathrm{ht}(T) \stackrel{\text{def}}{=} \min\{\alpha : \mathrm{Lev}_\alpha T = \varnothing\}.$$

Definition 7E. (Branch) A **branch** of tree T is a maximal chain (i.e. a maximal linearly ordered subset).

Example 7A. (Ordinals are trees) Each ordinal α can be seen as a tree with only one branch (α itself).

Example 7B. (The Cantor tree) The *Cantor tree* is ${}^{<\omega}2 \stackrel{\text{def}}{=} \bigcup_{n<\omega} {}^n2$ with the partial order $s \leqslant t$ iff $s \subseteq t$ (as functions). Recall that we can think of $s \in T$ as a finite sequence of 0s and 1s, so $s \leqslant t$ iff the sequence t extends s.

The Cantor tree has cardinality ω and height ω, and the nth level is exactly ${}^n2$. It also has 2^ω many branches, meaning that each branch can be identified by an infinite sequence over $\{0, 1\}$.

Example 7C. (The full Cantor tree) The *full Cantor tree* is $T = {}^{<\omega}2 \cup \{{}^\omega 2\}$ with the the partial order $s \leqslant t$ iff $s \subseteq t$.

One can define a topology on the full Cantor tree to make a separable space. This topology consists of every singleton $\{s\}$ for $s \in {}^{<\omega}2$, and a countable local basis around $t \in {}^\omega 2$ defined as, for $n \in \omega$,

$$U(t,n) = \{t \restriction m : m > n\}.$$

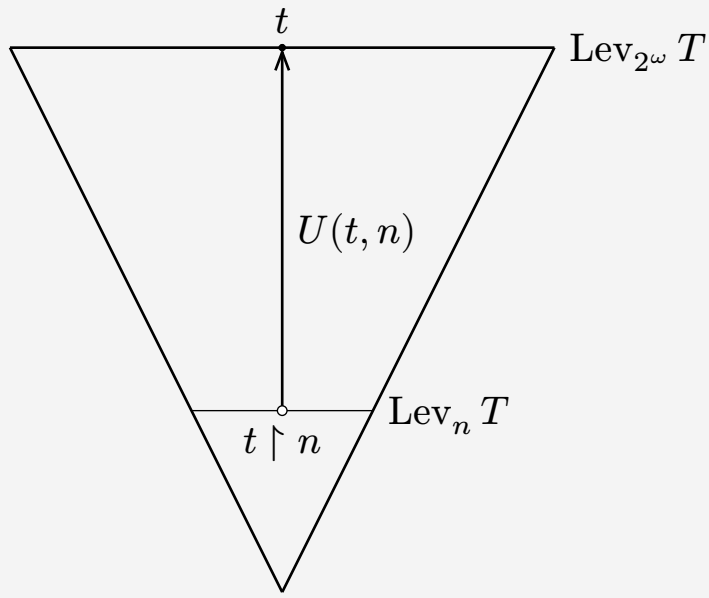


Figure 10: $U(t,n)$ is a basic open set of $t \in \mathrm{Lev}_{2^\omega} T$.

Definition 7F. (Antichain) An **antichain** of a tree T is a subset $A \subseteq T$ such that

$$\forall x, y \in A \quad x \neq y \Longrightarrow x \not\leqslant y \wedge y \not\leqslant x.$$

In other words, each distinct $x, y \in A$ are not comparable in $(T, \leqslant)$.

Lemma 7A. Each level of a tree is an antichain.

Proof. If $x \neq y \in \mathrm{Lev}_\alpha T$, then x and y are incomparable. Otherwise, WLOG suppose $x < y$, then the woset $\{t \in T : t < y\}$ would be order isomorphic to its initial segment, a contradiction. □

Definition 7G. A tree T is said to **split** $x \in T$ iff x has at least two immediate successors.

Definition 7H. For cardinal κ, a tree T is a **κ-tree** iff

$$\mathrm{ht}(T) = \kappa \text{ and } \forall \alpha < \kappa \ \ |\mathrm{Lev}_\alpha T| < \kappa.$$

Lemma 7B. (König) If T is an ω-tree, then T has an infinite branch.

Proof. We will construct a branch $\{x_n : n \in \omega\}$ recursively. To choose $x_0 \in T$, observe that T is a finite union of subtrees with some root $x_0 \in \mathrm{Lev}_0 T$,

$$T = \bigcup_{x_0 \in \mathrm{Lev}_0 T} \{t \in T : t \geqslant x_0\}.$$

So there must exist some $x_0 \in \mathrm{Lev}_0 T$ such that the subtree $T_0 = \{t \in T : t > x_0\}$ is infinite (ω is regular). Now, given that $\mathrm{Lev}_{n+1} T$ is finite, we can choose $x_{n+1} \in$

$\mathrm{Lev}_{n+1} T \subseteq T_n$ such that $T_{n+1} = \{t \in T : t > x_{n+1}\}$ is infinite. Finally, the set $\{x_n : n \in \omega\}$ forms an infinite branch in T. □

7.1 Aronszajn Trees

Definition 71A. An **Aronszajn** tree is an ω_1-tree in which every branch is countable.

In general, for a regular cardinal κ, a κ-Aronszajn tree is an κ-tree in which every branch has cardinality less than κ. König's lemma implies that there is no ω-Aronszajn tree. However, an (ω_1-)Aronszajn tree exists.

Theorem 71A. An Aronszajn tree exists.

Proof. We start with the following tree, ordered by inclusion (as functions):

$$S = \{s \in {}^{<\omega_1}\mathbb{Q} : s \text{ strictly increases}\}.$$

Thus S has height ω_1, since for any $\alpha < \omega_1$, there exists a strictly increasing $s : \alpha \hookrightarrow \omega \subseteq \mathbb{Q}$, in which case $s \in \mathrm{Lev}_\alpha S$. Each branch of S is also countable, because the union of a branch of S is an injection into the countable set $\mathbb{Q}$. However, S is not an Aronszajn tree. In particular, for any $\omega \leqslant \alpha < \omega_1$, the αth level has cardinality $2^\omega > \omega$.

This means that our tree has too many branches at levels ω or above, so we have to prune the tree. To do so, we will construct a subtree $T \subseteq S$ where each level $\alpha < \omega_1$ satisfies the following induction hypotheses:

(1) $|\mathrm{Lev}_\alpha T| < \omega_1$,
(2) $\mathrm{Lev}_\alpha T \subseteq \{s \in {}^{\alpha+1}\mathbb{Q} \subseteq S : s \text{ is continuous}\}$,
(3) $\forall \beta < \alpha\ \forall s \in \mathrm{Lev}_\beta T\ \forall q \in \mathbb{Q}\ \exists t \in \mathrm{Lev}_\alpha T\ (q > s(\beta) \to t > s \wedge t(\alpha) = q)$.

Intuitively, we want (1) each level to be countable, and (2) each sequence $s \in \mathrm{Lev}_\alpha T$ is continuous, meaning that for any limit ordinal $\beta < \operatorname{dom} s$, we have that

$$\sup_{\gamma<\beta} s(\gamma) = s(\beta).$$

Finally, we prune our tree with (3), which says that if $s \in T$ is a node below level α, and $q \in \mathbb{Q}$ is greater than its limit/maximum, then $\mathrm{Lev}_\alpha T$ has a sequence t that extends s and ends at q.

The construction of these levels consists of two stages, one for successor ordinals and one for limit ordinals. For successor ordinals, given the αth level, we define

$$\mathrm{Lev}_{\alpha+1} T = \bigcup_{s \in \mathrm{Lev}_\alpha T} \{t \in {}^{\alpha+2}\mathbb{Q} : t > s \text{ and } t(\alpha+1) > s(\alpha)\}.$$

Being a countable union of countable sets, (1) $\mathrm{Lev}_{\alpha+1} T$ is countable. At the successor stage, (2) is unaffected. Notice that (3) is satisfied when $\beta = \alpha < \alpha + 1$ by our choice of $t \in \mathrm{Lev}_{\alpha+1} T$. To see why (3) holds for $\beta < \alpha$ and $s \in \mathrm{Lev}_\beta T$, fix $q > s(\beta)$ and choose $r \in \mathbb{Q}$ such that $s(\beta) < r < q$ by density of $\mathbb{Q}$. Then apply (3) on $\mathrm{Lev}_\alpha T$, we obtain some $t' \in \mathrm{Lev}_\alpha T$ such that $t' > s$ and $t(\alpha) = r$. Finally, let $t = t' \cup \{(\alpha + 1, q)\} \in \mathrm{Lev}_{\alpha+1} T$, we have that $t > s$ and $t(\alpha + 1) = q$ as wanted (see Figure 11).

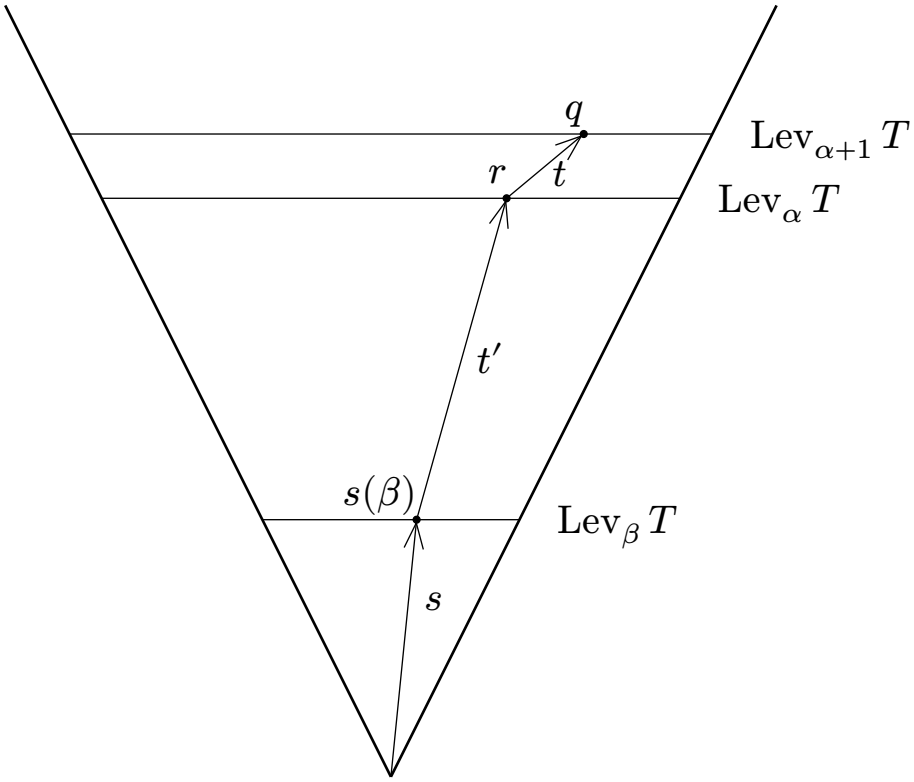


Figure 11: Construction of $\mathrm{Lev}_{\alpha+1} T$

Now assume α is a limit ordinal and given each $\mathrm{Lev}_\gamma T$ for $\gamma < \alpha$. We shall construct $\mathrm{Lev}_\alpha T$ where the most important step is to ensure continuity (2) at α. We first choose a sequence of ordinals (γ_n) such that

$$\gamma_0 < \gamma_1 < \gamma_2 < \cdots < \alpha \text{ and} \sup_{n<\omega} \gamma_n = \alpha.$$

Next, for each rationals $p < q$, we shall define $t_{p,q} : \alpha + 1 \to \mathbb{Q}$ as a node at level α. Let $p_0 = p$, we start by choosing an $s_0 \in \mathrm{Lev}_{\gamma_0} T$ such that $s_0(\gamma_0) = p_0$. To ensure continuity at α, fix a strictly increasing sequence of rationals p_n that converges to q and apply (3) on $\gamma_1, \gamma_2, \cdots$ to obtain $s_1, s_2, \cdots$ such that

$$s_0 < s_1 < s_2 < \cdots \text{ and } s_n(\gamma_n) = p_n.$$

Finally, define $t_{p,q} = \sup_{n<\omega} s_n \cup \{(\alpha, q)\}$ and our level is

$$\mathrm{Lev}_\alpha T = \left\{t_{p,q} : p, q \in \mathbb{Q} \wedge p < q\right\}.$$

Thus, (1) $|\operatorname{Lev}_\alpha T| \leqslant |\mathbb{Q} \times \mathbb{Q}| < \omega_1$ and (2) for $s \in \operatorname{Lev}_\alpha T$, we have that

$$\sup_{\gamma \leqslant \alpha} s(\gamma) = \sup_{n<\omega} s_n(\gamma_n) = s(\alpha).$$

And by our arbitrary choice of p and q, (3) is satisfied for $\operatorname{Lev}_\alpha T$ as well. □

7.2 The Suslin Hypothesis

Definition 72A. A **Suslin** line is a liearly ordered set $(X, <)$ such that in the order topology, X is ccc but not separable.

The **Suslin hyphothesis** (SH) says that "no Suslin line exists".

Lemma 72A. If Y is a Suslin line, then there exists a Suslin line X such that

1. X is dense in itself, i.e. $a < b \implies (a, b) \neq \varnothing$,
2. No nonempty open subset of X is separable

Proof. In Y, define equivalence relation $\sim$ where

$$x \sim y \iff (x, y) \text{ or } (y, x) \text{ is separable.}$$

Let $X = Y/\sim$ be the equivalence classes, note that each $I \in X$ is convex, meaning for $x, y \in I$

$$x < y \implies (x, y) \subseteq I.$$

We order X by $I < J$ iff $\exists x \in I \quad \exists y \in J \quad x < y$.

First, observe that each $I \in X$ is separable. Choose M to be a maximal disjoint family of nonempty open intervals in the form (x, y) for $x, y \in I \subseteq Y$. Since Y is ccc, M must be countable, so let $M = \{(x_n, y_n) : n < \omega\}$. Therefore, each $x_n \sim y_n$ (by convex), so let D_n be a countable dense subset of (x_n, y_n). Thus $D = \bigcup_n D_n$ is dense in $\bigcup_n (x_n, y_n)$, and we have that if $z \in I$ and $z \in (x, y) \subseteq I$, then (x, y) intersects some (x_n, y_n) by maximality, meaning $z \in \overline{D}$ unless $z = \min I$ or $\max I$. So $D \cup \{\min I, \max I\}$ is dense in I.

To show that X is dense in itself. BWOC assume there is $I, J \in X$ such that $I < J$ but (I, J) is empty. Since $I < J$, let $x \in I$ and $y \in J$ such that $x < y$. Therefore $(x, y) \subseteq I \cup J$ (convex) which is separable as observed before. Thus $x \sim y$, meaning (x, y) has a countable dense subset, contradicting that $(I, J) = \varnothing$.

To check the second condition, it suffices to show that for each $I < J$ in X, the interval (I, J) is not separable. BWOC let $\{K_n : 2 \leqslant n < \omega\}$ be a countable dense subset of (I, J), with $K_1 = I, K_2 = J$. Since each $K_n \in X$ is separable, let $D_n \subseteq Y$ be a countable dense subset of K_n. Now, $\bigcup_n D_n$ is dense in $\bigcup (I, J)$, meaning that points of I are equivalent to points of J (?), a contradiction.

Finally, to see that X is ccc, BWOC assume $\{(I_\alpha, J_\alpha) : \alpha < \omega_1\}$ is pairwise disjoint collection of nonempty intervals. Thus pick $x_\alpha \in I_\alpha$ and $y_\alpha \in J_\alpha$ for $x_\alpha < y_\alpha$, we have that $\{(x_\alpha, y_\alpha) : \alpha < \omega_1\}$ is also pairwise disjoint collection of nonempty intervals, contradicting that Y is ccc. □

Lemma 72B. If X is a Suslin line, then X^2 is NOT ccc

Proof. By indunction on $\alpha < \omega_1$, we wish to find $a_\alpha, b_\alpha, c_\alpha \in X$ such that

1. $a_\alpha < b_\alpha < c_\alpha$,
2. (a_α, b_α) and (b_α, c_α) are nonempty,
3. (a_α, c_α) does not contain any b_ξ for $\xi < \alpha$.

Assuming we have chosen these points, let $U_\alpha = (a_\alpha, b_\alpha) \times (b_\alpha, c_\alpha)$. 2. guarantees that U_α is nonempty, and 3. shows that each U_α are pairwise disjoint. To see the latter, let $\xi < \alpha$, thus $b_\xi \notin (a_\alpha, c_\alpha)$. If $b_\xi \leqslant a_\alpha$, then (a_ξ, b_ξ) and (a_α, b_α) are disjoint. Otherwise, $c_\alpha \leqslant b_\xi$, so (b_ξ, c_ξ) and (b_α, c_α) are disjoint.

Now we proceed to find these points. Let W be the set of isolated points of X. Since singletons of the isolated points form a family of pairwise disjoint nonempty open sets, $W \subseteq X$ is countable. By induction, assuming we have chosen a_ξ, b_ξ, c_ξ for $\xi < \alpha$, we first note that $X - \overline{W \cup \{b_\xi : \xi < \alpha\}}$ is a nonempty open set (otherwise $W \cup \{b_\xi : \xi < \alpha\}$ is a countable dense subset of X). Therefore it contains a basic open set (a_α, c_α). Since this interval contains no isolated points, it must be infinite, thus there exists some $b_\alpha \in (a_\alpha, c_\alpha)$ such that (a_α, b_α) and (b_α, c_α) are nonempty. □

Corollary 72C. $\mathsf{MA}(\omega_1) \vdash \mathsf{SH}$

Now we move our focus back to trees, and establish an

> It is considered good practice to prune a tree, removing branches that are sickly or do not product fruit. At the same time, we may ensure that the tree has a single trunk.
>
> — Kunen, 1980

Definition 72B. A **well-pruned** κ-tree is a rooted κ-tree T such that

$$\forall x \in T \quad \forall \alpha < \kappa \quad \exists y \in \operatorname{Lev}_\alpha T \quad \operatorname{ht}_T(x) < \alpha \to x < y.$$

That is, given $x \in T$, then every level above $\operatorname{ht}_T(x)$ has a successor of x.

The following lemma allows us to work with only well-pruned trees without loss of generality.

Lemma 72D. For any regular κ, if T is a κ-tree, then T has a well-pruned κ-subtree.

Proof. Define a κ-subtree S by the following condition:

$$x \in S \iff |\{z \in T : x < z\}| = \kappa$$

To show its well-pruned (not necessary rooted yet), consider any $x \in S$ and α such that $h_S(x) < \alpha < \kappa$. Then define

$$\begin{aligned} Y &= \{y \in \operatorname{Lev}_\alpha T : x < y\} \in [T]^{<\kappa}, \\ Z &= \bigcup_{\beta > \alpha} \{z \in \operatorname{Lev}_\beta T : x < z\}. \end{aligned}$$

Intuitively, Y consists of all successors of x at αth level, and Z are the successors above αth level. Now, since T is a κ tree where κ is regular, $|T - Z| < \kappa$, so $|Z| = \kappa$. Then, because each $z \in Z$ is the successor of some $y \in Y$, we can rewrite Z as

$$Z = \bigcup_{y \in Y} \{z \in Z : y < z\}.$$

Therefore, there must exist some $y_0 \in Y$ for which the set $\{z \in Z : y_0 < z\}$ has size κ. This establishes that $y_0 \in \operatorname{Lev}_\alpha S$. A similar argument using the observation that T is the union of $\{y \in T : y > x_0\}$ for some $x_0 \in \operatorname{Lev}_0 T$ where $|T| = \kappa$ is regular shows that $\operatorname{Lev}_0 S$ nonempty.

Finally, choose an $x_0 \in \operatorname{Lev}_0 S$, the set $\{y \in S : y \geqslant x_0\}$ is a well-pruned subtree of T. □

Lemma 72E. For regular κ, if T is a well-pruned κ-Aronszajn tree, and $x \in T$, then

$$\forall n < \omega \quad \exists \alpha > \operatorname{ht}_T(x) \quad |\{y \in \operatorname{Lev}_\alpha T : y > x\}| \geqslant n$$

Proof. For $n = 2$, note that $\{y : y > x\}$ meets all levels above x, but cannot form a chain. For $n > 2$, we proceed with induction. Assuming the lemma for n, let $\alpha > \operatorname{ht}_T(x)$ and $y_1, \cdots, y_n \in \operatorname{Lev}_\alpha T$ such that each $y_i > x$. Now, let $\beta > \alpha$ such that there exist $z_n, z_{n+1} \in \operatorname{Lev}_\beta T$ above y_n. Then, the well-prunedness implies that for each $1 \leqslant i < n$, there is $z_i \in \operatorname{Lev}_\beta T$ above y_i. Thus, $\{z_1, \cdots, z_n, z_{n+1}\}$ is the desired set. □

Definition 72C. A **Suslin tree** is T such that $|T| = \omega_1$ and every chain, antichain are countable.

Remark 72A. A Susline tree is a Aronszajn tree.

Theorem 72F. There is a Suslin tree iff there is a Suslin line

Proof. ($\Longrightarrow$) Let T be a Susline tree, WLOG assume it is well-pruned.

Let $L = \{B \subseteq T : B \text{ is a branch}\}$. Thus, for each $B \in L$, there exists an ordinal $h(B)$ such that $h(B)$ contains exactly one node from $\operatorname{Lev}_\alpha T$ for $\alpha < h(B)$, and no element from $\operatorname{Lev}_\alpha T$ for $\alpha \geqslant h(B)$. For each $\alpha < h(B)$, let $B(\alpha)$ be the element in $B \cap \operatorname{Lev}_\alpha T$. Since T is an Aronszajn tree, $h(B)$ is countable, and since T is well-pruned, $h(B)$ has no maximum element, thus a limit ordinal.

Next, we endow L with a linear order $<_L$ to make it a Suslin line. For distinct $A, B \in L$, define

$$d(A, B) = \min\{\alpha : A(\alpha) \neq B(\alpha)\}.$$

Note that $d(A, B) < \min\{h(A), h(B)\}$. Now fix a linear order $<_T$ on T, and define, for $A, B \in L$

$$A <_L B \Longleftrightarrow A(d(A, B)) <_T B(d(A, B)).$$

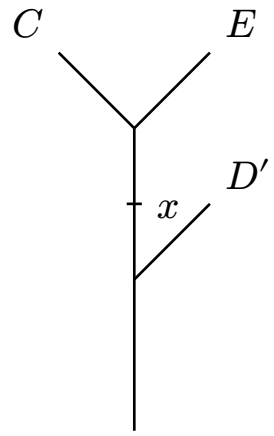
To show L is ccc. BWOC let $\{(A_\xi, B_\xi) : \xi < \omega_1\}$ be a family of pairwise disjoint nonempty open intervals. For each ξ, pick $C_\xi \in (A_\xi, B_\xi)$ and ordinal $\alpha_\xi < h(C_\xi)$ such that (recall that $h(C_\xi)$ is a limit ordinal)

$$\min\{d(A_\xi, C_\xi), d(C_\xi, B_\xi)\} < \alpha_\xi$$

Therefore, $\{C_\xi(\alpha_\xi) : \xi < \omega_1\}$ form an antichain in $(T, <)$, contradicting that T is Suslin.

To show L is not separable, it suffices to show that for each $\delta < \omega_1$, the set $\{B : h(B) < \delta\}$ is not dense. Fix $x \in \operatorname{Lev}_\delta T$ and let $\alpha > \delta$ such that $y, z, w \in \operatorname{Lev}_\alpha T$ are distinct successors of x. Pick branches $C, D, E \in L$ that contains y, z, w respectively, and WLOG assume $C <_L D <_L E$. Thus (C, E) is a nonempty open interval that does not intersect $\{B : h(B) < \delta\}$. To see why, consider any $D' \in (C, E)$ and BWOC assume $d(C, D') < \operatorname{ht}(x)$. Since $x \in C \cap E$, we must have that $d(C, D') = d(D', E) < \operatorname{ht}(x)$, thus $C(d(C, D')) = E(d(D', E))$, contradicting that (notice strict order)

$$C(d(C, D')) <_T D'(d(C, D')) = D'(d(D', E)) <_T E(d(D', E)).$$



□

Proof. ($\Longleftarrow$) Let $(L, <_L)$ be a Suslin line. Let $(\mathcal{U}, <)$ be the collection of all nonempty open intervals of L, ordered by reverse inclusion:

$$I < J \Longleftrightarrow J \subset I.$$

We proceed to build $T \subseteq \mathcal{U}$ such that $(T, <)$ is a Suslin tree. To do so, we shall build each level as $\mathcal{U}_\beta$ for $\beta < \omega_1$ satisfying

1. $\mathcal{U}_\beta$ consists of pairwise disjoint open intervals;
2. $\bigcup \mathcal{U}_\beta$ is dense in L;
3. For any $\alpha < \beta$, any $I_\alpha \in \mathcal{U}_\alpha$, any $I_\beta \in \mathcal{U}_\beta$,

$$I_\alpha \cap I_\beta = \varnothing \vee \left(I_\alpha \leqslant I_\beta \wedge I_\alpha - \overline{I_\beta} \neq \varnothing\right).$$

Assume such $\mathcal{U}_\beta$'s exist, we let $T = \bigcup_\beta \mathcal{U}_\beta$ and thus T is tree with $\operatorname{Lev}_\beta T = \mathcal{U}_\beta$. If $A \subseteq T$ is an antichain, then A consists of pairwise disjoint open intervals, which must be countable because L is ccc.

If $\left\{I_\xi : \xi < \omega_1\right\}$ is a (increasing) chain, then for each $\xi < \eta$ we have

$$I_\xi \leqslant I_\eta \text{ and } I_\xi - \overline{I_\eta} \neq \varnothing$$

meaning that $\left\{I_\xi - \overline{I_{\xi+1}} : \xi < \omega_1\right\}$ contradicts that L is ccc.

Finally $|T| = \omega_1$ because density of $\bigcup \mathcal{U}_\beta$ implies $\operatorname{Lev}_\beta T = \mathcal{U}_\beta \neq \varnothing$.

Now we shall construct each level. Choose $\mathcal{U}_0$ to be a maximal disjoint subfamily of $\mathcal{U}$; note thatthe maximality ensures that its union is dense.

Given $\mathcal{U}_\alpha$, we construct $\mathcal{U}_{\alpha+1}$ as follows: for each $I \in \mathcal{U}_\alpha$, let K_I be a maximal disjoint subfamily of

$$\left\{K \in \mathcal{U} : I \leqslant K \wedge I - \overline{K} \neq \varnothing\right\}.$$

Therefore, $\mathcal{U}_{\alpha+1} = \bigcup_{I \in \mathcal{U}_\alpha} K_I$ does the job.

Now let γ be a limit ordinal and assume $\mathcal{U}_\alpha$ has been constructed for $\alpha < \gamma$. Let

$$\mathcal{K} = \left\{K \in \mathcal{U} : \forall \alpha < \gamma \quad \forall I \in \mathcal{U}_\alpha \quad I \cap K = \varnothing \vee \left(I \leqslant K \wedge I - \overline{K} \neq \varnothing\right)\right\}$$

Let $\mathcal{U}_\gamma$ be a maximal disjoint subfamily of $\mathcal{K}$, thus 1. and 3. holds. To see why 2. hold, it suffices to show that no $J \in \mathcal{U}$ is disjoint from all members of $\mathcal{U}_\gamma$, which is also implied by

$$\forall J \in \mathcal{U} \quad \exists K \in \mathcal{K} \quad K \subseteq J$$

given the maximality.

To begin, let E be the set of all left and right endpoints of all intervals in $\bigcup_{\alpha<\gamma} \mathcal{U}_\alpha$. Therefore E is countable, whereas J is not separable. Fix $K_1 \in \mathcal{U}$ such that $K_1 \subseteq J$ and $K_1 \cap E = \varnothing$. Thus K_1 doesn't contain any endpoint of any $I \in \bigcup_{\alpha<\gamma} \mathcal{U}_\alpha$, meaning $I \cap K_1 = \varnothing$ or $K_1 \subseteq I$. Finally, let $K \in \mathcal{U}$ such that $K_1 \leqslant K$ and $K_1 - \overline{K} \neq \varnothing$, we have that $K \subseteq J$ and $K \in \mathcal{K}$. □

8 Čech–Stone Compactification

Definition 8A. Let $\omega^* \stackrel{\text{def}}{=} \{p : p \text{ is an ultrafilter on } \omega\}$, then $\beta\omega \stackrel{\text{def}}{=} \omega^* \cup \omega$ is the topological space generated by the basis that consists of

1. $W(A) \stackrel{\text{def}}{=} \{p \in \omega^* : A \in p\}$ for any $A \subseteq \omega$,
2. $\{n\}$ for any $n \in \omega$.

Definition 8B. A topological space is **separable** iff it contains a countable dense subset.

Remark 8A. No space with an uncountable discrete subspace is separable.

Definition 8C. A topological space is **sequentially compact** iff every sequence has a convergent subsequence.

Remark 8B. $\beta\omega$ is compact but not sequentially compact as no sequence converges.

Remark 8C. ω_1 endowed with the order topology is sequentially compact. Since every sequence in ω_1 is bounded, we can extract an increasing subsequence whose limit is the supremum.

Definition 8D. For $p \in \omega^*$, a point $x \in \beta\omega$ is a p-**limit** of a sequence $(x_n \in \beta\omega)$ iff $\forall U \in \eta(x) \quad \{n \in \omega : x_n \in U\} \in p$.

Remark 8D. p-limits are unique in a T_2 space.

Exercises

Exercise 8.1. Find the p-limits of sequences in the discrete subspace ω of $\beta\omega$.

Exercise 8.2. Show that $\beta\omega$ is p-compact for all $p \in \omega^*$